



**Verified Carbon
Standard**

TABREED DISTRICT COOLING GROUP PROJECT



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Summary:**A description of the validation and the project**

Validation: National Central Cooling Company PJSC - Tabreed has appointed KBS Certification Services Limited to do validation of the VCS project activity “Tabreed District Cooling Group Project”, with regard to the relevant requirements of VCS Standard version 4.5.

Project: The project activity “Tabreed District Cooling Group Project” is a grouped project which employs CDM methodology AM0117: Introduction of a new district cooling system — Version 2.0./40/. The project involves installing new district cooling plants of various capacities to supply cooling to residential and commercial consumers in Abu Dhabi, United Arab Emirates. The Project will replace isolated and less efficient cooling technologies (such as air-cooled reciprocating chiller systems) and thereby reducing the consumption of electricity and corresponding Carbon Dioxide (CO₂) emissions.

The purpose and scope of validation

Purpose: The purpose of a validation is to have a thorough and independent assessment of the proposed project activity against the applicable VCS requirements, in particular, the project's baseline, monitoring plan and compliance with the relevant VCS and host Party criteria. These are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of emission reductions. KBS's objective is to perform a thorough, independent assessment of the validation of the project activity.

Scope: Validation scope is defined as an independent and objective review of the Project Description (PD). The PD is reviewed against the relevant criteria and guidance documents provided by VCS which include the following: VCS Standard, version 4.5 Updated 04-October-2023 /43 a/, VCS Program Guide, version 4.4 Updated 29-August-2023 /43 b/, VCS Validation and Verification Manual, version 3.2 Updated 19-October-2016 /43 c/, Registration & Issuance Process, version 4.4 Updated 04-October-2023 /43 d/, VCS Program Definitions version, version 4.4 Updated 29-August-2023 /43 e/, applicable at the time in order to confirm that the project meets the applicability conditions of the selected baseline and monitoring VCS methodology AM0117 (version 2.0), also assess the claims and assumptions made in the PD without limitation on the information provided by the project participants.

The method and criteria used for validation

The validation consists of the following four phases:

- I. A desk review of the project description documents
 - A review of data and information;
 - Cross checks between information provided in PD and information from sources with all necessary means without limitations to the information provided by the project proponent;

II. On-site interviews with project stakeholders

- Interviews with relevant stakeholders in host country with personnel having knowledge with the project development via telephone, email or direct on-site visits;
- Cross checking between information provided by interviewed personnel with all necessary means without limitations to the information provided by the project proponent;

III. Reference to available information relating to projects or technologies similar to project under validation and review based on the approved methodology being applied for the appropriateness of formulae and accuracy of calculations.**IV. The resolution of outstanding issues and the issuance of the final validation report and opinion.****The number of findings raised during validation**

During the course of validation, a total of 10 findings were raised, which include:

03 Corrective Action Requests (CARs);

04 Clarification Requests (CLs);

03 Forward Action requests (FARs).

All the raised CAR and CL have been successfully closed by the Project Proponent.

Any uncertainties associated with the validation

There are no uncertainties associated with the validation of the project activity. The validation has been done with a reasonable level of assurance.

Summary of the validation conclusion

KBS Certification concludes the validation with a positive opinion that the VCS Grouped Project “Tabreed District Cooling Group Project” as described in the PD (version 1.9, dated 10-May-2024) /1.2/, meets all applicable VCS requirements, including those specified in the VCS Standard (v4.5, updated 04-October-2023), relevant methodology, tools and guidelines.

The selected baseline and monitoring methodology (AM0117 version 2.0) are applicable to the project and correctly applied. KBS Certification therefore requests the registration of the project as a VCS grouped project.

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1 INTRODUCTION

1.1 Objective

National Central Cooling Company PJSC - Tabreed has commissioned KBS Certification Services Limited to do VCS validation of the project activity: "Tabreed District Cooling Group Project", with regard to the relevant VCS requirements of VCS Standard version 4.5; to attain real, measurable, additional and permanent emission reductions.

The purpose of validation is to ensure a thorough, independent assessment of the project description (PD), in particular the project's description eligibility, ownership, crediting period and details regarding its compliance with the law statutes and other regulatory frameworks.

- The requirements of VCS Program Guide Version 4.4 /43b/
- The requirements of VCS Standard Version 4.5 /43a/
- To assess the project's compliance with other relevant rules, including the project country (United Arab Emirates) legislation and
- Other relevant rules are validated in order to confirm that the project description as documented is sound and reasonable and meet the stated requirements and identified criteria.

The validation is seen as necessary to provide assurance about the quality of the project and its intended generation of emission reductions over the project's crediting period without any double counting.

1.2 Scope and Criteria

The scope of the validation is defined as an independent and objective review of the Project description document, where methodological deviations and the project's compliance (i.e., Project Details) is reviewed against the requirements of VCS Standard. KBS has employed a risk-based approach in the validation, focusing on the identification of significant risks for project implementation and the generation of Emission Reductions.

The items covered in the validation are described below:

- VCS Criteria, VCS standard version 4.5
- VCS Project Description
- Background investigation and follow up interviews
- Draft validation reporting with CARs, CLs & FARs, if any

- Final validation reporting

The validation is based on the information made available to KBS and on the contract conditions.

The validation is not meant to provide any consulting to the project proponent. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project description. The work carried out by KBS is free from any conflict of interest.

1.3 Reasonableness of Assumptions

- ☒ Reasonable level of assurance

The validation report is based on VCS-PD, supporting documents made available to the validator and information collected through performing interviews. Based on the process and procedures conducted, KBS states whether the information in the PD:

- is materially correct and is a fair representation of the actual project details, and
- is prepared in accordance with VCS requirements and the applied methodology for information pertaining to description, methodological deviations and reporting.

Through the on-site interview, host country rule and regulations related to project activity, Project description, technological measures, Implementation, Operation, Management of project activity and Training of personnel, Stakeholder consultation etc. has been checked and found appropriate.

The validation work is carried out as per this requirement and the validation opinion is assured provided the credibility of all above. Details are presented in the Validation statement in section 4 below.

1.4 Summary Description of the Project

TABREED DISTRICT COOLING GROUP PROJECT (hereafter called “Project” or “Grouped Project”) is implemented by National Central Cooling Company PJSC (TABREED). It is a grouped project, which will install district cooling plants of various capacities to meet the cooling demand for residential and commercial consumers in the Abu Dhabi Emirates, United Arab Emirates (UAE).

From the baseline survey report /47/, the validation team noted that the project activity will displace the cooling demand supply from conventional/less efficient cooling system and reduce the electricity consumption which would result in reduction in greenhouse gas (GHG) emissions. The validation team verified the plant design documentation /14/, technical specifications of the project equipments /04/ to conform the technology implemented in the project activity.

The purpose of the Project instance 1 is to introduce new district cooling plant with a prospect for installing thermal energy storage system (in future) to supply cooling demand to a commercial consumer through a dedicated distribution network- The Sea World, Yas Island Abu Dhabi, United

Arab Emirates. the validation verified the Diagram for cooling demand distribution network from DCP in project Instance 1 to Sea world /59/, wherein the validation team could evidence that a new dedicated distribution network is installed from the new DCP to the sea world building. To further corroborate that the DCP in the project activity serves only the new building, sea world, the validation team also cross checked the Quantity of cooling demand supplied (measured at DCP) (5377481 TRH) and the quantity of cooling demand consumed (measured at sea world) – (5362750 TRH); the quantities are found to be matching with a minor loss (less than 0.3f % owing to line loss).

The first project activity instance is installed in the sea world Abu Dhabi with a capacity of 9,000 TR. The estimates of annual average GHG emission reductions of project activity instance 1 is 19,320 tCO₂e/annum. The PP has submitted separate ER calculation sheet /2.2/. The calculations are found to be correctly determined.

2 VALIDATION PROCESS

2.1 Method and Criteria

The Validation process was carried out in line with the requirements of VCS Standard version 4.5. The project activity applies CDM approved methodology, AM0117, Version 2.0 /40/, categorized under sectoral scope 1 “Energy Industries (renewable/non-renewable”. The validation consisted of the following phases:

- Completeness check and desk review of the Project Description
- Interview with project representatives and issuance of findings.
- Resolution of findings followed by preparation of the final validation report and opinion.

Timeline of Validation:

Work order signed	03-June-2023
Onsite Inspection	25-september-2023.
(Draft) Reporting	20-October-2023
(Final) Reporting	20-May-2024

2.2 Document Review

After the submission of the draft PD /1.1/ and supporting background documents related to the project design and baseline from the client, the completeness of information made available is reviewed as per VCS standard version 4.5 /43a/ requirements. Furthermore, the validation team

used additional documentation by third parties like host party legislation, technical reports referring to the project design, baseline technologies or to the basic conditions and technical data available on public domain. The project has been listed in the VERRA (<https://registry.terra.org/app/projectDetail/VCS/4519>). A desk review is carried out to assess the following:

- the project details as per VCS PD template
- any methodological deviations applied
- compliance with relevant law and regulations
- conditions prior to project initiation/ baseline

The list of documents reviewed is included in the section 'References'.

2.3 Interviews

Please refer section 2.4, where complete list of interviewed personnel and key points discussed is provided.

2.4 Site Visits

The VVB has conducted the on-site visit on 25-September-2023 by following all the Covid-19 Travel guidelines notified by Verra & in accordance with local site requirements. As per the news published by Verra on "Verra to Resume Project Site Visits" dated 12-August-2022: <https://verra.org/verra-to-resume-project-site-visits>, the VVB has notified the Verra in advance regarding its conduct of site visit by filling the "Notice of Validation/Verification Services(NOV) form" /55/.

Details of interviewees, topics covered and additional information are presented below:

Date:	<i>25-September-2023</i>		
Key points discussed:	Name of person, interviewed	Designation, Organization	Team member
Host Country rule and regulations related to project activity, Project description, Ownership, Project start date, technology, Conditions prior to the Project initiation,	Tushar Joshi	Assistant Manager – Mechanical (TABREED)	Team Leader: Mr. Srinivasan Selvaraj Technical Expert (1.1): Mr. Sauvik Banerjee)
	Shabareesh	Electrical Inspector - TABREED	
	Omar Majdalawi	Project Inspector - TABREED	

Participation under other GHG program, procedure to avoid double counting, monitoring plan	Pradeep	Supervisor	
	Avinash	Operator, TABREED	
	Aziz Ullah	Operator, SCECO	
	Shashi Prakash	Head Regulation, MASDAR	

2.5 Resolution of Findings

KBS applies the risk-based approach aimed at focusing on high-risk issues to the validation results whilst not omitting any part of the mandatory processes. A few discrepancies were found during the validation and the findings was submitted to the project proponent, indicated under the titles corrective action requests (CARs) and clarification requests (CLs). CARs and CLs require the PP to take relevant actions. Criteria for judging items as CAR or CL are as follows:

Corrective action request (CAR):

- the project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions
- the Voluntary Carbon Standard's requirements have not been met, or
- there is a risk that emission reductions cannot be monitored or calculated.

Clarification request (CL):

- Information is insufficient or not sufficiently clear to determine whether the applicable VCS requirements have been met.

FAR is to be raised to highlight issues related to project implementation that require review during the first verification of the project activity. FARs does not relate to VCS requirements for registration.

CARs and CLs are to be resolved or closed out if the PP modifies the project description, rectifies the PD or provides adequate additional explanations or evidence that satisfies the concerns. If this is not completed, the project activity cannot be recommended for registration under VCS registry.

03 CARs, 04 CLs and 3 FARs were found during validation and closed satisfactorily. The list of CARs/CLs raised and the response provided, the means of validation, reasons for their closure and references to correction in the VCS PD are provided in Appendix 3 of this report. The revised PD with changes incorporated as per the issues raised were rechecked with the documentary evidences and found to be inline.

2.5.1 Forward Action Requests

03 Forward Action Requests are raised during the validation and is to be checked during next periodic verification.

As a result, the VCS PD that was listed initially /1.1/ underwent key changes as follows:

- a. Project capacity is changed as per the capacity, implemented for the Phase 1 of project instance 1.

b. Baseline scenario has been accurately determined in accordance with the requirements set out in the applied methodology.

c. Additionality is demonstrated with supporting document, as per Para 5.2.1 of the methodology: Step 1 Penetration and performance assessment

d. Eligibility criteria are defined more appropriately for inclusion of new project instances.

e. Appropriate references are now included for all the assumptions in the Emission reduction calculation.

f. Monitoring and measurement methods and procedures are incorporated as per the requirements of the applied methodology and tools.

The changes are incorporated correctly in the revised VCS PD /1.2/.

3 VALIDATION FINDINGS

3.1 Project Details

National Central Cooling Company PJSC (TABREED) is implementing TABREED DISTRICT COOLING GROUP PROJECT (hereafter called “Project” or “Grouped Project”). The objective of the project is to reduce fossil-fuel based electricity consumption by providing efficient district cooling solution (air conditioning) to meet the cooling demand for residential and commercial consumers in the Abu Dhabi, United Arab Emirates (UAE). The validation team confirmed the project type and technology from the Environmental permit, issued by Environmental Agency – Abu Dhabi (Licence No. CE-237) /21/ and from the District cooling plant design documentation /14/. The validation team noted that the district cooling system will displace the cooling from conventional/less efficient cooling system and reduce the electricity consumption which would result in reduction in greenhouse gas (GHG) emissions.

Based on interview with the PO, the validation team noted that in the UAE air conditioning is necessary to survive, because of extreme weather conditions and hence, the penetration of cooling technology is almost 100%. Most of the residential and commercial consumers use window air conditioner or split air conditioner to meet the cooling demand. The validation team also noted that the commercial consumers have started shifting towards air cooled centralized cooling, which are less efficient with respect to district cooling.

The purpose of the Project is to introduce new district cooling plant with a state-of-the-art thermal energy storage system to supply cooling to residential and commercial consumers in the Sea World, Yas Island Abu Dhabi, United Arab Emirates. The Project will replace isolated and less efficient cooling technologies (air-cooled reciprocating chiller systems) and thereby reducing the consumption of electricity and corresponding Carbon Dioxide (CO₂) emissions. The PP has estimated that the district cooling will save more than 40-50% of electricity with respect to the baseline technology. The assessment details of this estimation is provided in section 3.3.6.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Audit history	The first type of audit, the project validation, is being carried out by KBS Certification Services Limited. The PP has signed a contract with KBS for this service /57/. The validation report dated 23-February-2024 is being submitted by KBS certification to VERRA for registration of the project activity.
Sectoral scope	The project provides district cooling to commercial consumers through a dedicated cooling distribution network (Energy efficiency improvement project). Hence the choice of sectoral scope: 01 Energy industries (renewable - / non-renewable sources) is deemed to be appropriate.
AFOLU project category, if applicable	Not Applicable
Project activity type	The project is a Grouped Project activity with each project instances aiming to provide district cooling to commercial consumers through a dedicated cooling distribution network (Energy efficiency improvement project). Hence the choice of project activity type is deemed to be appropriate.
General eligibility of the project to participate in the VCS Program	- The project does not belong to the projects excluded in Table 1 of VCS Standard 4.5. Thus, the project is eligible under the scope of the VCS program. - The project was listed on the project pipeline on 21-August-2023 and the opening meeting between VVB and the PP was conducted on 25-September-2023. The validation team checked and confirmed that the project is listed on the project pipeline as under validation before conducting the opening meeting. Hence, the project meets the requirements set out in section 3 of the registration and Issuance process, v4.4 /43d/. The start date of the project activity is 25-May-2023. Hence, as per para 3.8.1 of the VCS standard v4.5, the validation shall be completed within two years of the project start date i.e., before 25-May-2025. The validation is now completed and validation report dated 23-february-2024 is being submitted for registration with Verra. - The project applies CDM approved methodology AM 0117, which was inactivated on 23-September-

	2023. The validation team noted that as Per the VCS Standard, v4.5 (PDF), Section 3.22.3(4), Verra has set the grace period for inactivated CDM methodologies to twelve months after the date the methodology was inactivated. Project proponents that have already requested listing on the Verra Registry before the inactivation date may complete validation within twelve months after the inactivation date. The inactivation date for this methodology was Sep 23, 2023 and therefore the validation team noted that the project can be submitted for registration with Verra until 22-September-2024 Hence the validation team concludes that the project is eligible to apply the methodology, AM0117 v2 under the VCS program until 22-September-2024. The project is a grouped project with each project instances aiming to provide district cooling to residential and commercial consumers through a dedicated cooling distribution network The project is not a fragmented part of a larger project or activity that would otherwise exceed such limits.
AFOLU project eligibility, if applicable	Not applicable, as the project is not AFOLU project.
Transfer project eligibility, if applicable	Not applicable, as the project is not a transfer project.
Project design	This is a grouped project, with each project instance aiming to install district cooling plants of various capacities to residential and commercial consumers who would have used the less efficient air-cooled reciprocating chiller system. Each new project instance will be confirmed against the eligibility criteria below. The eligibility criteria have been provided clearly in section 1.5.1 of the VCS PD /1.2/ and then justification provided for inclusion of project activity instances. The first project activity instance is installed in the sea world Abu Dhabi with a capacity of 9,000 TR.

No.	Eligibility criterion - Category	Eligibility criterion - Required condition	Supporting evidence for inclusion	VVB Assessment
1	<i>Applicability Condition of the applied methodology.</i>	The project activity instances shall meet applicability conditions for applicable methodology (AM 0117 version 02) as defined in section 2.2.	The justification shall be part of 'Monitoring Report' during the first-time inclusion of any new instance.	For the first instance of the group project, the PP has demonstrated compliance to methodology applicability conditions in section 3.2 of VCS PD /1.2/. The justification shall be part of 'Monitoring Report' during the first-time inclusion of any new instance.
2	Geographical Area	The project activity instances for the grouped project will be located in the geographic boundary of Abu Dhabi Emirates, and owned by the project owner or the subsidiary of the project owner.	KML file, GPS Coordinates, maps as per technical reports, proposals and feasibility studies, Sales documents, commissioning and approval documents.	Through the on-site interviews, it is confirmed that the geographical boundary included is Abu Dhabi Emirates and is consistent with the geographical boundary in the project Description.
3	Baseline scenario	All Project Activity Instances	Latest/updated third party report	PP has demonstrated the baseline

		shall meet the baseline definition as defined in the respective methodology. The baseline scenario is based on technology penetration.	shall be provided for every new project instance inclusion.	scenario for the first project instance in line with the requirements provided in para 5.2 in the applied methodology /40/. Thus, the eligibility criteria has been met for the new project activity instance under this group project. The demonstration shall be part of 'Monitoring Report' during the first-time inclusion of any new instance.
	4	<i>Technology type:</i> The project activity instances to be included in the grouped project activity is the implementation of new district cooling plant with or without thermal energy storage, which will	Technical reports, proposals and feasibility studies Cooling demand Lay-out diagram for the DCP. Confirmation by PP regarding applicable Project Scenario for the Project activity.	As confirmed through the on-site interview and through review of project description and specifications /04/, the technology deployed under the project activity shall be District Cooling Plant (in compliance with the methodological requirements)/ 40/.

		be servicing the new development/buildings	Purchase order of equipment. Declaration from Project Instance operator to the PP. Evidence on start of operation of the connected buildings.	
	5	Additionality	The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology. Additionality for this instance has been demonstrated by using requirements provided in para 5.2.1; Step 1: Penetration and Performan The seasonal energy efficiency ratio (SEER) calculations. The share of technologies can be derived from official country reports, third party surveys and/or credible international sources (e.g. International Energy Agency).	Through the on-site interview and through review of third party survey report /12/, the validation team noted that the project instance is not a common practice and hence is additional in line with the requirements provided in para 5.2.1; Step 1: Penetration and Performance assessment, in the applied methodology /40/. Thus, the eligibility criteria has

			ce assessment, in the applied methodology.		been met for the new project activity instance under this group project.
			a. The seasonal energy efficiency ratio (SEER) of the DCP involved in the project activity is higher than a benchmark SEER and (b) The share of the district cooling technologies at the moment of the project registration is less than 20 per cent of all cooling technologies within the benchmark boundary in terms of cooling output.		
			The future instances shall comply above		

		requirement of methodology for additionality demonstration.		
	6	Start date	The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.	The evidence to support the start date of future instances shall be furnished to VVB. FAR 01 is raised in this regard.
	7	Capacity Limits	The validation team noted that no capacity limits are mentioned in the applied methodology and therefore no capacity	Not applicable

			limit will be applied for inclusion of new project activity instances.		
	8	Stakeholder consultation:	The local stakeholder consultation will be conducted before inclusion of any new project instance.	Meeting notices; Minutes of meeting; Documents relating to Feedback / comments redressal.	During the on-site interviews and through the review of local stakeholder consultation report /30c/, it has been confirmed that PP has conducted the Local stakeholder consultation. Thus, the eligibility criteria has been met for the first project activity instance under this group project.
	9	Environmental Impact:	Every new project instance will follow the host country's environmental regulations related with environmental approvals	Environmental Impact study, if applicable.	For the first project instance, there is no requirement from the host country to conduct environmental impact assessment. For future instances, the environmental impact study will be conducted, following the

				host country's regulations.
10	Sampling :	No sampling is used for the emission reduction calculation . All the project instances shall have the monitoring parameters as explained in monitoring section of the methodology AM0117 version 02.	Not applicable	Not applicable
11	Conditions to avoid double counting - <i>Unique Identification</i> :	All the new project instances will be named uniquely based on location as the PI 1 is called 'Sea World District Cooling Plant'. To prevent double counting of District Cooling Plants under the project, District	PI title, GPS coordinates, technical reports, proposals and feasibility studies. Analysis of projects in VCS/CDM pipeline.	Through the On-site interviews and review of Technical reports /04/, it is confirmed that the unique marking of the DCP will be done by user's information, collected through the Contract and shall be compiled in project records. Project records shall include username, identification number and address/locatio

			cooling plant shall be uniquely marked, and new project instance implementers shall provide assurance s/declarations that they are not included in any other CDM or Voluntary GHG offset schemes outside of this project		n of the DCP, Geo coordinates of DCP and commissioning date. The record keeping system will ensure that each DCP can be traced to avoid double counting. In addition, the PP shall submit a declaration confirming that the DCP implemented in the new project instances are not part of any other CDM or Voluntary GHG trading scheme.
	12	<i>Program me management- Approval of new project instance by project proponent; collection of data and record keeping:</i>	The project proponent approves each project instance to be included into its registered group project activity. Project proponent Tabreed will be responsible for keeping all the emission	Statement of project proponent giving approval for the new project instance to be included into its registered group project activity.	The validation team noted that for the first instance, project proponent is the implementer; hence no further approval is required, however for the cases, where implementer other than PP, an approval on the same shall be given by PP in order to include the instance into

		reduction data, including the details of project instances start date, commissioning document and other required information.		this grouped project.
	13	Condition s to provide an affirmati on that funding from Annex I parties, if any, does not result in a diversion of official developm ent assistanc e.	under this project, new project instance operators shall be required to provide an affirmation that funding from annex I parties, if any, does not result in a diversion of official developme nt assistance	Individual new project instance operators shall sign ODA declarations for new project instances implemente d by them to attest that no funding from annex 1 parties that leads to diversion of ODA has been used in the project. the statement shall specify whether or not annex 1 funding is used within the project where funding is sought from annex 1 parties the

			ODA declaration shall be supported by details regarding annex 1 funding that affirms that the funding doesn't divert ODA. For this instance, no funding from Annex-1 parties is received declaration to this effect has been furnished to VVB.	
	14	VCU ownership	End users receiving Cooling demand under the specific new project instance contractually cede their rights to claim and own emission reductions under the VCS to the project proponent of the group project.	A default distribution agreement for end users including the provision that emission reductions generated by the DCP are owned by the project proponent, will be provided for each new project instance. Sample agreement with end user for During the on-site interview and through the review of documents viz., communication agreement with VERRA /54/, Sample beneficiary agreement with end users /29/, the validation team noted that the PP for the project is TABREED. The sample agreement submitted to the validation team contained a provision which states that emission

			Waiver of carbon credits are submitted to the VVB for the first project instance.	reductions generated by the DCP are owned by the project proponents which shall give the ownership of VCU to the PP. This will be verified during the verification when end user list is available for new project instances.
	15	Awareness and agreement of those operating a new project instance on group project subscription	Contractual provisions to ensure that those operating the new project instance are aware and have agreed that their activity is being subscribed to the new project instance.	Evidence for inclusion, in case new project instance operators are different from the project proponent: a declaration from project proponent/implementer, stating that they are aware and have agreed that their activity is being subscribed to the group project will be provided for each new project instance. The validation team noted from interview with the PP that for the first instance, project proponent is the only implementer; Hence, no further requirements related to awareness and agreement on their instance to be subscribed under this grouped project, is required to be furnished. However, for the cases, where implementer other than PP, an agreement on the awareness between PP and

					new instance implementer shall be furnished. Thus, the eligibility criteria has been met for the new project activity instances under this group project.
	The eligibility criteria for new instances of grouped projects conforms with the VCS Program requirements.				
Project ownership	NATIONAL CENTRAL COOLING COMPANY PJSC – TABREED is the Project Proponent (PP), which is as per the submitted communication agreement with VERRA /54/. The project activity has been developed by NATIONAL CENTRAL COOLING COMPANY PJSC – TABREED. Therefore, project proponent has sole right to use the generated Emission reductions. Validation team confirms that the Chief Operating Officer (Mr. Antonio Di Cecca), as mentioned in the section 1.8 of the VCS-PD, participated in the on-site assessment conducted on 25-September-2023. The project commissioning certificate for project activity /05/ is the supporting documents to demonstrate the project ownership. This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.5) “a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals”. From the Musataha Agreement between Miral Asset Management LLC (MAM) and National Central Cooling Company PJSC (TABREED), in respect of a District cooling plant site at the SeaWorld Abu Dhabi Development on Yas Island, Abu Dhabi /08/, the validation team noted that MAM owns rights to the site and has agreed to grant to TABREED rights in respect of the plot from 30-August-2022 to 30-August-2052 /08/. Also from section 2.8 of this agreement, the validation team noted that the TABREED owns the District Cooling plant.				

	The PP has stated in the VCS PD that during the registration of DCP for project instances, the end user will sign a declaration to transfer the ownership rights of the carbon assets generated from this project to the project proponent; the validation team confirmed the same from the sample beneficiary agreement submitted by the PP /29/. However, the project ownership for each project instance, (satisfying the requirements of Section 3.7 of VCS Standard version 4.5), is to be verified from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions). Hence a FAR 01 is raised in this regard.
Project start date	The project start date is 25-May-2023. In accordance with section 3.8 of VCS standard, Version 4.5 /43a/, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or removals. Accordingly, the start date of the project is considered as 25-May-2023 which is the date of commissioning of the first project instance, which was confirmed by verifying the commissioning certificate /05/. Thus, the validation team confirms that the start date of the project activity is reasonably determined.
Project crediting period	☒ Seven years, twice renewable <i>25-May-2023 to 24-May-2030</i> The VCS Standard version 4.5 /43a/ section 3.9.1 states that, “For non-AFOLU projects, the project crediting period shall be either seven years, twice renewable for a total of 21 years, or ten years fixed”. PP has chosen crediting period of 7 years, twice renewable for a total of 21 years. Hence, the first crediting period starts from 27-May-2023 and ends on 26-May-2030 (both days included). Thus, the validation team confirms that the project crediting period is reasonably determined.
Project scale	The estimated volume of emission reduction from the project activity is 19,320 tons of CO₂e per year, which is lower than 300,000 tons of CO₂e per year. Hence, the validation team confirms that the project activity falls under the category 1 ‘Projects’ as per para 3.10.1 of the VCS standard version 4.5

	/43a/, as it has the potential to reduce GHG emission less than 300,000 tCO ₂ e/year.
Likelihood of achieving estimated GHG emission reduction or removals	The estimated total CO ₂ e emission reduction by the project activity over the crediting period of 7 years are expected to be 135,242 tCO ₂ e respectively. The validation team confirmed that emission reduction forecast is reasonable if the underlying assumptions do not change.
Technologies and measures implemented by the project activity or activities	National Central Cooling Company PJSC (TABREED) is implementing TABREED DISTRICT COOLING GROUP PROJECT. The objective of the project is to reduce fossil-fuel based electricity consumption by providing efficient district cooling solution (air conditioning) to meet the cooling demand for residential and commercial consumers in the Abu Dhabi, United Arab Emirates (UAE). The validation team confirmed the project type and technology from the Environmental permit, issued by Environmental Agency – Abu Dhabi (License No. CE-237) /21/ and from the District cooling plant design documentation /14/. The validation team noted that the district cooling system will displace the cooling from conventional/less efficient cooling system and reduce the electricity consumption which would result in reduction in greenhouse gas (GHG) emissions.
Implementation schedule of the project activity or activities	The project is already implemented at the time of validation. The project was commissioned on 25-May-2023 and has been operating since then. The validation team verified the commissioning certificate /05/ to confirm the commissioning date of the project.
Project location	The grouped project will be hosted across Abu Dhabi Emirates and will cover the geographical boundary of the Abu Dhabi Emirates. The project boundary is defined by the following GPS coordinates and is found to be consistent. Further, project GPS coordinates as provided in the VCS PD and which are presented below, have been verified through the means of google earth software. Latitude: 22° 40' N to 25° North Longitude: 51° to 56° East

	The validation team noted that the project activity instance 1 is installed in Sea World Abu Dhabi with following coordinates: <table border="1" data-bbox="764 327 1369 758"> <tr> <td>Facility Name</td><td>Sea World District Cooling Plant</td></tr> <tr> <td>Facility Activity</td><td>District Cooling Plant</td></tr> <tr> <td>Plot Area</td><td>10099.95-meter square</td></tr> <tr> <td>Location</td><td>Yas Island Sector YS6</td></tr> <tr> <td>Plot number</td><td>P10</td></tr> <tr> <td>Latitude</td><td>24.483103977 N</td></tr> <tr> <td>Longitude</td><td>54.618530045 E</td></tr> </table> The validation team confirmed the location details from the Environmental Permit application submitted to the Environmental Agency – Abu Dhabi (EAD) /22/ and the Environmental Permit granted by the Agency /21/. The location details were also confirmed during the on-site visit to the plant.	Facility Name	Sea World District Cooling Plant	Facility Activity	District Cooling Plant	Plot Area	10099.95-meter square	Location	Yas Island Sector YS6	Plot number	P10	Latitude	24.483103977 N	Longitude	54.618530045 E
Facility Name	Sea World District Cooling Plant														
Facility Activity	District Cooling Plant														
Plot Area	10099.95-meter square														
Location	Yas Island Sector YS6														
Plot number	P10														
Latitude	24.483103977 N														
Longitude	54.618530045 E														
Conditions prior to project initiation	The baseline scenarios are the same as the conditions existing prior to the project initiation. As per the methodology AM 0117 the baseline scenario is the continuation of the cooling energy production by the baseline cooling technologies. The assessment of conditions prior to project initiation is provided in section 3.3.4 of this report.														
Project compliance with applicable laws, statutes and other regulatory frameworks	The validation team noted that the project is a voluntary effort by the project proponent. The validation team confirms that the Project complies with all applicable laws. The main laws are: Environmental laws of Abu Dhabi: The project has already received the environmental permit for construction and operation /21/. District cooling regulatory framework under Department of Energy (DoE)¹: Under Law No 11 of 2018, the Department of Energy (DoE) has a duty to regulate the central / district														

¹ <https://www.doe.gov.ae/en/Legislation-and-Compliance/Economic-Regulations/District-Cooling>

	cooling sector in the Emirate of Abu Dhabi. Following extensive consultation with the public and private stakeholders, the DOE has launched the District Cooling regulations upon the government approval, comprising technical, legal and economic dimensions. The economic dimension of this framework will look into aspects of market competition and price regulation, to ensure that: • There is increased fairness and transparency in the market as well as enhance service levels for end-users. • The price for new district cooling schemes is fair and lower than the cost of the best alternative available for cooling. • The benefits of district cooling are properly reflected in end-users tariffs. • District cooling providers can recover their efficient costs and sector profitability is fostered. • The price control process is easy to implement and sustainable in the long term. The two technical regulations have been published and project is in compliance with these codes: District Cooling Technical Code dated 15/03/2020 /09a/ District Cooling Metering Code dated 15/03/2020 /09b/ The validation team has also confirmed that the baseline scenario presented in the VCS PD/1.2/ is also in compliance with the laws and regulations of the host country. Validation team confirm that the project is a voluntary initiative and is not under any mandatory/compliance scheme of the host country. Net GHG emission reductions or removals generated by this PA instance will not be used for compliance with any emissions trading program or to meet binding limits on GHG emissions as the host country i.e. UAE is not a participant in any emission trading programs or nor does it have any binding limits.
Double counting and participation under other GHG programs	The PP has confirmed in the section 1.16 of the PD that the grouped Project had not been registered or currently seeking registration under any other GHG programs. The project has not participated under any other GHG program. The grouped project or any of the project activity instances included in it

	shall neither seek nor apply for any other form of greenhouse gas -related environmental credit, including renewable energy certificates. The PP has submitted a Declaration in this regard for not availing other forms of environmental credit for the same crediting period under consideration, /20/. The validation team has verified project portals from other programs including CDM and confirmed that this grouped Project had not been registered or currently seeking registration under any other GHG programs. The same was discussed and confirmed with the PP and found to be correct, hence accepted.
No double claiming with emissions trading programs or binding emission limits	The project does not reduce GHG emissions from activities that are included in an emissions trading program or binding emission limit or any other mechanism that includes GHG allowance trading.
No double claiming with other forms of environmental credit	The project has not sought or received another form of GHG-related credit, including renewable energy certificates. The PP has submitted a declaration in the regard /20/.
Supply chain (Scope 3) emissions double claiming	The validation team noted that the Project participant does not buy or sell any product whose emission footprint is changed by the project activity. Further there is no scope 3 emissions claimed by the project participant. The validation team, from interviewing the PP, noted that the PP has not issued any public statement on this issue.
Sustainable development contributions	The proposed Project Activity contributes to sustainable development in the United Arab Emirates by the following means: Economic indicators The Project Instance contributes to the 1. Conservation of energy sources through a more efficient way of generation and distribution of cooling energy, thus allowing for decreased electricity consumption per ton of cooling supplied. This contributes to UN SDG indicator 11 'sustainable cities' Target 11.6: "Reduce the environmental impacts of cities". 2. Education and training of staff in the operation of the district cooling system. In addition, staff will be trained in the application of the international VCS standards of the Project Instance Activity monitoring and quality assurance / quality control. Social indicators

	The Project Instance contributes to the positive quality of life and benefits the local community by helping ensure a less energy intensive generation and supply of cooling energy. Environmental indicators The Project Instance 1. Will be reducing an estimated total of 19,320 tCO₂ of emission reductions per year and therefore contributing to UN SDG 13 'Climate Action'. The validation team has verified the emission reduction calculation sheet for the first project instance /2.2/. 2. Complies with all applicable national environmental regulations and standards and has no impact on air, water, soil or groundwater pollution. The validation team has verified the environmental permit, issued by Environmental Agency – Abu Dhabi (Licence No. CE-237) dated 09-March-2023 to the proponent, National Center Cooling Company PJSC for construction and operation /21/. 3. will be reducing the usage of desalinated water as it allows for the future introduction of grey water (treated sewage water) to be used as cooling medium. Technical indicators The Project Instance introduces a new and energy efficient technology and know-how into the United Arab Emirates, which can successfully be replicated to other cities within the region. Based on interview with the PP, the validation team noted that the Project Instance will contribute to the increase in utilization and application of know-how of this type of technology (district cooling).
Additional information relevant to the project	- Leakage Management for AFOLU Projects No leakage management as the project activity is not an AFOLU project. - Commercially Sensitive Information All the information required for the VCS validation of this project activity has been provided by the PP and thus, there is no commercially sensitive information in this project activity, which has been excluded from the public version of the project description.

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	The validation team noted that the PP has announced the concept of the project through media releases /30a/ in January 2021 i.e., before the implementation of the project. The PP has also provided options for the stakeholders to submit any feedback/grievances through the contact details available in the company website. Based on interview with the PP, the validation team noted that the PP has adopted established procedures for stakeholder identification. Relevant stakeholders are found identified – Employees, Government authorities including Abu Dhabi Distribution Company, Environmental Agency-Abu Dhabi, Department of Energy, NGOs and environmental organizations who are affected or likely to be affected by project activities.
Legal or customary tenure/access rights	The PP has obtained access rights to territories and resources from the relevant stakeholders. From the Musataha Agreement between Miral Asset Management LLC (MAM) and National Central Cooling Company PJSC (TABREED), in respect of a District cooling plant site at the SeaWorld Abu Dhabi Development on Yas Island, Abu Dhabi /08/, the validation team noted that MAM owns rights to the site and has agreed to grant to TABREED rights in respect of the plot from 30-August-2022 to 30-August-2052 /08/. Also from section 2.8 of this agreement, the validation team noted that the TABREED owns the District Cooling plant.
Stakeholder diversity and changes over time	The validation team noted that the PP has identified stakeholders from diverse field such as government authorities, local community, NGOs and employees. No changes are expected over time as the site of project is already a developed plot in YAS Island. Moreover, the validation team noted that the PP has implemented a mechanism for

	ongoing communication with the stakeholders, through which all stakeholders can request for redress of any grievance.
Expected changes in well-being	The Project will replace isolated and less efficient cooling technologies (such as air-cooled reciprocating chiller systems) and thereby reducing the consumption of electricity and corresponding Carbon Dioxide (CO ₂) emissions. Hence, validation team noted that there is no change in well-being and other stakeholder characteristics relative to the baseline scenario as the cooling output is same in both the scenarios (Project and baseline).
Location of stakeholders	All the stakeholders, interested parties and Local community are located within the Emirate of Abu Dhabi. The validation team noted that since the project impacts are limited to Emirate of Abu Dhabi, the location of the identified stakeholders are deemed appropriate.
Location of resources	The validation team noted that MAM owns rights to the project site and has agreed to grant to TABREED rights in respect of the plot from 30-August-2022 to 30-August-2052 /08/.

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	The PP has provided a description of local stakeholder consultation process in Section 2.1.2 of the VCS PD. The validation team noted that the PP has announced the concept of the project through media releases /30a/ in January 2021 i.e., before the implementation of the project. The PP has also provided options for the stakeholders to submit any feedback/grievances through the contact details available in the company website. The validation team noted that the PP has not received any comments relating to the project instance. Based on interview with the PP /44/, the validation team noted that there is no legal requirement for the PP to conduct a local stakeholder meeting for the project activity. However, since the PP wished to pursue the project for VCS registration, the PP conducted local stakeholder consultation meeting in-order to meet the requirements

	of VCS standard. This issue was also raised during the VERRA review during the Pipeline listing of the PD /45/. Hence, the PP conducted another Local stakeholder meeting on 19-September-2023 with an objective to provide information about the project in an accessible manner to all the relevant stakeholders. The number of stakeholders who attended this meeting is 20. The validation team noted that the PP has adopted established procedures for carrying out effective stakeholder consultation meeting, as described below: i) Relevant stakeholders are found identified – Employees, Government authorities including Abu Dhabi Distribution Company, Environmental Agency-Abu Dhabi, Department of Energy, NGOs and environmental organizations who are affected or likely to be affected by project activities; ii) Appropriate time and place are identified. The validation team noted that the PP has informed the place of meeting– online meeting and time of the meeting atleast 10 days in advance. iii) Appropriate means of outreach is used. The PP has sent invitations to the identified stakeholders through an email on 06-September-2023. The validation team noted that the PP has invited the stakeholders in advance. iv) The PP has made a power point presentation about the project during the meeting. The PP has submitted a copy of the PPT to the validation team. It could be evidenced that PP has included all relevant information about the project.
Consultation outcome	From reviewing the minutes of this meeting, the validation team noted that the PP has given opportunities to the participants of the meeting, to invite comments and feedback and to answer any questions raised. The validation team noted that the PP has not received any negative comments relating to the project instance.
Ongoing communication	The validation team noted that PP has established a customer service charter (https://www.tabreed.ae/customer/) which provides the detailed process on how the complaints are recorded and addressed.

Stakeholder input	The validation team noted that the PP has not received any negative comments relating to the project instance. The validation team noted from the minutes of the meeting /30c/, that the PP has responded to the questions/clarifications about the project. Also the validation team noted that PP has implemented on-going communication channels through which the stakeholder inputs are obtained on a continuous basis and the comments will be recorded and appropriately responded.
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3.2.1.2 Free, Prior and informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	Project has received all the required consents from government authorities. The validation team confirms that the Project complies with all applicable laws. The main laws are: Environmental laws of Abu Dhabi: The project has already received the environmental permit for construction and operation /21/.
Outcome of FPIC discussion	Project has received all the required consents from government authorities and other interested parties. The project has received the environmental permit for construction and operation /21/. The project has not encroached on land, relocated people without consent and forced physical or economic displacement. The validation team noted that MAM owns rights to the site and has agreed to grant to TABREED rights in respect of the plot from 30-August-2022 to 30-August-2052 /08/. Also from section 2.8 of this agreement, the validation team noted that the TABREED owns the District Cooling plant.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	The validation team noted that PP has established a customer service charter (https://www.tabreed.ae/customer/) which provides the detailed process on how the complaints are recorded and addressed.
Grievance redress procedure	The validation team noted from interview with the PP that the PP has established a dedicated portal for stakeholders (end users) to register their grievances. The PP informed that End-user Customers can lodge complaints by calling 800 TASLEEM (8275336/8227333), emailing customerservice@tasleem.ae, or visiting the Tasleem website: www.tasleem.ae. The validation team noted that these channels are working at the time of validation site visit.

3.2.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
No comments received during the Public commenting period from 21-August-2023 to 20-September-2023.	No comments received during the Public commenting period.	Validation team confirms that the project was listed on VCS website https://registry.terra.org/app/projectDetail/VCS/4519 and was open for public comments from 21-August-2023 to 20-September-2023. No comments were received during the 30 days' period.
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3.2.2 Respect for Human Rights and Equity

3.2.2.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination and sexual harassment	The validation team noted that the Project proponent, Tabreed was Established in 1998 and is a leading company providing sustainable, energy efficient and innovative cooling solutions to government, commercial, residential and private organizations in the GCC countries. The PP, during the interview, mentioned that the company abides by all the applicable law in UAE. The validation team noted that in UAE labour laws prohibit any type of discrimination and harassment. Federal Decree-Law No. 33/2021, known as the Labour law, expressly addresses discrimination in various aspects, emphasizing the importance of treating employees fairly and equally. Article 4 of the Labour Code prohibits discrimination based on race, colour, sex, religion, national origin, social origin, or disability.
Management experience	The Validation team noted that the project proponent, Tabreed, was Established in 1998 and is a leading company providing sustainable, energy efficient and innovative cooling solutions to government, commercial, residential and private organizations in the GCC countries. Hence the validation team noted that the PP has experience in similar projects in last 25 years and have sufficient experience on managing all the requirements.
Gender equity in labor and work	The validation team, during the interview with the PP, noted that Tabreed has established procedures for gender equality and it publicly publishes the ESG related information. The validation team noted from the report available in the following link that the PP provides equal opportunities in the context of gender equity and pay for labor and work. https://www.tabreed.ae/wp-content/uploads/2023/07/Tabreed_ESG_Report_2022.pdf
Human trafficking, forced labor, and child labor	The validation during the interview with the PP, noted that the PP abides the law in UAE, which prohibit any type of human trafficking, forced labor and child labor.

3.2.2.2 Human Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Human rights	The validation team, during interview with the PP, noted from ESG report 2022 /56/ that Tabreed is committed to adhering to international laws and conventions, including, but not limited to, the International Labour Organization (ILO) core conventions: Conventions No. 29 and No. 105 on the Abolition of Forced Labour; Conventions No. 138 and No. 182 on the Abolition of Child Labour; and Conventions No. 100 and No. 111 on Non-discrimination.
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3.2.2.3 Indigenous Peoples and Cultural Heritage

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Preservation and protection of cultural heritage	The validation team noted from interview with the PP that there is no cultural heritage site near the project site. The project has received all the approvals for the structures of the building from the respective authorities. The authorities while granting approval verifies whether the project preserves and protects cultural heritage. Also, Hence the validation team concludes that the project follows all the required regulations.

3.2.2.4 Property Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Rights to territories and resources	Project owner has all the rights on the project. The land has been leased and the construction has been carried out by project owner. The validation team noted that MAM owns rights to the site and has agreed to grant to TABREED rights in respect of the plot from 30-August-2022 to 30-August-2052 /08/. Also from section 2.8 of this agreement, the validation team noted that the TABREED owns the District Cooling plant.
Respect for property rights	During interview with the PP, the PP informed that the Tabreed is established in 1998 and that during this journey, it has respected the rights of other stakeholders, as mentioned in the agreement terms.

3.2.2.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	Not applicable as the benefits from VCU issuance are not shared with any other stakeholder. NATIONAL CENTRAL COOLING COMPANY PJSC – TABREED is the Project Proponent (PP), which is as per the submitted communication agreement with VERRA /54/. The project activity has been developed by NATIONAL CENTRAL COOLING COMPANY PJSC – TABREED. Therefore, project proponent has sole right to use the generated Emission reductions.
Summary of the benefit sharing plan	Not applicable
Approval and dissemination of benefit sharing plan	Not applicable.

3.2.3 Risks to Local Stakeholder and the Environment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Risks to stakeholder participation	Project is energy efficiency and operation is limited to expert operators and therefore no risk to stakeholder participation.
Working conditions	The project owner is listed public company and abide by all the legal requirements.
Safety of women and girls	The service provided by project is common to all irrespective of gender and therefore no risk identified.

Safety of minority and marginalized groups, including children	The service provided by project is common to all irrespective of gender and therefore no risk identified.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	The validation team noted that the project is housed in a dedicated building. The PP informed that the PP monitors air, noise, discharge to water, generation of waste, release of hazardous materials regularly as per the requirements in the environmental permit/21/. The limits are adhered.

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	The project will not adversely impact biodiversity and ecosystems. During interview with the project proponent, the validation team noted the project site was already a developed land and no impact on biodiversity due to project.
Soil degradation and soil erosion	During interview with the project proponent, the validation team noted the project site was already a developed land and no soil degradation and soil erosion is envisaged.
Water consumption and stress	During the project site visit, the validation team noted that the project consumes sea water from sea water desalination plant. There is no impact envisaged for potable water.
Usage of fertilizers	The project does not use any fertilizer.

3.2.4.1 Rare, Threatened and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Species and habitat	The project will not adversely impact habitats for rare, threatened, or endangered species. During site visit, the validation team noted that the project is not located in or adjacent to habitat for rare, threatened or endangered species.
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3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
<i>Nil</i>	Not applicable as the project does not involve any introduction of species. It involves installing new district cooling plants of various capacities to supply cooling to residential and commercial consumers.

3.2.4.3 Ecosystem Conversion

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
<i>Nil</i>	Not applicable as the project does not involve any ecosystem conversion. It involves installing new district cooling plants of various capacities to supply cooling to residential and commercial consumers.

3.3 Application of Methodology

3.3.1 Title and Reference

As evident from the UNFCCC website, the applied methodology is approved and the applied version is valid. The applied methodology and the applied tools are listed below.

Methodology: AM0117: Introduction of a new district cooling system --- Version 2.0.

Tools applied:

1. Tool for the demonstration and assessment of additionality (version 07)
2. Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion (version 03)
3. Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03)
4. Tool to calculate the emission factor for an electricity system (version 07)

5. Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period (version 03.0.1)
6. Additionality of first-of-its-kind project activities (version 03.0)

3.3.2 Applicability

Compliance with each applicability condition as listed in the chosen baseline and monitoring methodology AMO117 (version 02) and the applicable tools has been demonstrated.

The assessment was carried out for each applicability criterion and the details are presented below:

Methodology ID	Applicability condition	Assessment and conclusion
AM0117 (version 02)	The methodology is applicable to project activities that reduce CO₂ emissions by means of one, or a combination, of the following measures: (a) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through a new dedicated distribution network; (b) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through an existing dedicated distribution network; (c) Expansion of the existing district cooling system(s) by adding a new district cooling plant(s) with or without expanding	Applicable The validation team noted that the Project activity Instance 1 is new district cooling system that supply cooling to commercial consumers through dedicated distribution network. The same was confirmed by reviewing the purchase order issued for the key project equipments /10/, technical specifications of the key project equipments /04/, Approvals/Clearances from Government authorities /11/ corroborated during the on-site visit.. For project instance 1, the validation team verified the distribution network diagram /59/ (The same diagram provided in Section 1.12 of revised VCS PD), wherein the validation team could evidence that a new dedicated distribution network is installed from the new DCP to the sea world building. To further corroborate that the DCP in the project activity serves only the new building, sea world, the validation team also cross checked the Quantity of cooling demand supplied (measured at DCP) and the quantity of cooling demand consumed (measured at sea

	a dedicated distribution network.	world); the quantity are found to be matching. Hence, this criteria is met.
AM0117 (version 02)	Emission reductions that are gained due to the switch of the energy sources shall not be claimed by applying this methodology alone. Therefore, emission reductions due to displacement of the baseline power source can be claimed by the means of the application of this methodology in combination with another relevant approved methodology. In doing so, interactive effects shall be considered as per the “Guidelines for the consideration of interactive effects for the application of multiple CDM methodologies for a programme of activities”. For example, if the project district cooling plant is powered partially or completely by a dedicated renewable energy power plant, a project proponent may wish to consider small scale methodology ‘AMS-I.F: Renewable electricity generation for captive use and mini-grid’ to account for emission reductions due to shift of the power source. Otherwise, the most conservative electricity emission factor should be applied to determine emission reductions.	Not applicable The validation team noted that the Project does not replace the energy source. It remains the electricity from the grid. It was confirmed from the grid connection evidenced during the site assessment. The project does not have facility for any other source of electricity to run the chillers. Therefore this condition is not applicable.

Tool	Applicability Condition	Assessment and conclusion
Tool 01: Tool for the demonstration and assessment of additionality (version 07)	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The Project is demonstrating additionality as per the procedures provided in the approved methodology. Hence this criteria is not applicable
	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Not applicable The PP has used provision from the applied Methodology to demonstrate additionality. Hence this criteria is not applicable
Tool 03: ‘Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion	This tool provides procedures to calculate project and/or leakage CO ₂ emissions from the combustion of fossil fuels. It can be used in cases where CO ₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	Applicable The project involves calculation of CO ₂ emissions from fossil fuel combustion based on the quantity of fuel combusted and its properties. The PP has included formula with respect to the determination. Hence this criteria is not applicable
Tool 05 ‘Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption:	Applicable The project consume electricity from grid. The electricity is purchased from the grid. The validation team

	(a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	confirmed the same from verifying the grid connection and through receipts of electricity purchases /53/. There is no fossil fuel captive power plant at site. Hence the project activity meets the conditions described in scenario A regarding source of electricity consumption.
Tool 07 'Tool to calculate the emission factor for an electricity system'	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand side energy efficiency projects).	Not Applicable Due to unavailability of the data for calculation of OM, BM and CM, project proponent has requested for a deviation to use publicly available grid emission factor from government sources, for the project. The assessment of the deviation is described in

		detail in section 3.3.7 of this report.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off grid power plants. In the latter case, two sub options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off grid power generation” should be met. Namely, the total capacity of off grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The project activity consumes electricity from UAE National grid and hence, the emission factor for the project electricity system, used in the project activity refers to grid power plants only.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The validation team noted that the project electricity system is totally in United Arab Emirates (UAE), a Non-Annex 1 Country. As evident from the reports provided by UAE’s national power utility /48/, UAE does not import electricity from other countries. Hence, the tool is applicable.
	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	No biofuel is used in the project activity. Hence this criteria is not applicable.

Tool 11 "Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period"	This tool provides a stepwise procedure to assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period, as required by paragraph 49 (a) of the modalities and procedures of the clean development mechanism. The tool consists of two steps. The first step provides an approach to evaluate whether the current baseline is still valid for the next crediting period. The second step provides an approach to update the baseline in case that the current baseline is not valid anymore for the next crediting period.	Applicable The Project will use this tool during renewal of crediting period.
Tool 23:" Additionality of first-of-its-kind project Activities"	This methodological tool is applicable to project activities that wish to use the "first-of-itskind" approach to demonstrate additionality and that use versions of baseline and monitoring methodologies, or the "Tool for the demonstration and assessment of additionality" or the "Combined tool to identify the baseline scenario and demonstrate additionality", which allow using the "first-of-its-kind" approach for demonstrating additionality.	Not applicable The Project is demonstrating additionality as per the procedures provided in the approved methodology. It is not using first-of-its-kind approach for the additionality demonstration and therefore this tool is not applicable.

Thus the validation team confirms that the project participant correctly applied the approved CDM methodology AM0117, version 02 /40/ to the project activity. Based on the document review and on-site visit, VVB hereby confirms that the selected baseline and monitoring methodology is applicable to the Project, which complies with all the applicability conditions therein and the selected version is valid at the time of submission of the proposed project activity for registration. It is also confirmed that the methodology is correctly applied by comparing it with the actual text of the applicable version of the methodology.

3.3.3 Project Boundary

The project boundary was checked with regard to the paragraph 12 and 13 of the methodology /40/. Boundaries (geographically and related to GHG sources / sinks) have been correctly given in PD. The details of the assessment are provided below:

Para 24	Spatial Extent of project boundary	Assessment and conclusion
a	The District Cooling System, including district Cooling plant (s), pipes, Substations and buildings, existing and new that are or will be connected to the district cooling system;	The project boundary comprises Sea World district cooling system, including district cooling plant(s) pipes, sub-stations and buildings, new, that are or will be connected to the district cooling system. This was confirmed during the site assessment for the project instance 1. For all future instances, this will be verified at the time of inclusion.
b	The captive power plant/unit and/or all power plants/units connected physically to the electricity system that the CDM project activity is connected to.	The project boundary comprises all power plants connected physically to the UAE electricity system. This was confirmed during the site assessment for the project instance 1. For all future instances, this will be verified at the time of inclusion.

All sources and GHGs are included in the project boundary as required in the applied methodology. The details of the assessment are provided in the table below. The provisions of the applied Methodology and tools were crosschecked against the PD. Based on review of the documents such as commissioning report for the project instance 1 /05/ and observations of the site and equipment used in the process /44/, it could be confirmed that all sources and GHGs required by the methodology have been included within the project boundary and clearly indicated in the PD. An appropriate justification has been provided in the PD. Justification is reasonable and is in line with information provided in supporting documentation.

Source		Gas	Included?	Assessment and conclusion
Baseline	Electricity and/or thermal energy consumed by baseline cooling technologies	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification, this is conservative
		N ₂ O	No	Excluded for simplification, this is conservative
Project	Electricity and/or thermal	CO ₂	Yes	Main source of emissions
		CH ₄	No	Excluded for simplification

Source		Gas	Included?	Assessment and conclusion
	energy consumed by the district cooling system	N ₂ O	No	Excluded for simplification

3.3.4 Baseline Scenario

The description of baseline identification in the PD is transparent and verifiable. The baseline scenario has been identified through the application of the steps prescribed by the applied methodology to be ‘continuation of the cooling energy production by the baseline cooling technologies’.

The assessment of compliance to the requirements of step-wise procedure is explained below:

Paragraph number/ Condition	Assessment of VVB
14. The methodology assumes that the baseline scenario is the continuation of the cooling energy production by the baseline cooling technologies.	Applicable In Project the baseline scenario is continuation of the cooling energy production by baseline technologies.
15. District cooling projects implemented in least developed countries (LDCs) are deemed to be automatically additional	Not Applicable UAE is non- LDC and therefore automatic additionality is not applicable.
16. If district cooling projects are implemented in non-LDCs and face the first-if-its-kind barrier, the latest approved version of the “Additionality of first-	Applicable UAE is non – LDC and this option is applicable. However, the Project is not applying this additionality demonstration option.

of-its-kind project activities” shall be followed to demonstrate the additionality of these project activities.	
17. For district cooling projects, which are implemented in non-LDCs and which are not the first-of-its-kind, the procedure described below shall be applied.	Applicable The Project is using this option for additionality and baseline justification.
5.2.1. Step 1: Penetration and performance assessment 18. This step aims to determine whether the proposed project activity is common practice in the city where the proposed CDM project activity is intended to be implemented.	Applicable The Project activity is not a common practice in the UAE. Through the on-site interview and through review of third party survey report /12/, the validation team noted that the project instance is not a common practice and hence is additional in line with the requirements provided in para 5.2.1; Step 1: Penetration and Performance assessment, in the applied methodology /40/. The validation team noted that as per para 19 (b), the share of technologies can be derived from official country reports, third party surveys and/or credible international sources (e.g. International Energy Agency). The project owner has appointed a third Party, Global Market insights, to carry out the Abu Dhabi cooling sector analysis to provide the benchmark and penetration of cooling technologies, based on which the penetration rate of district cooling technology is determined as 18%. The validation team has crosschecked the penetration rate through the government publication ‘ Abu Dhabi Energy Outlook 2050 (https://www.doe.gov.ae/-/media/Project/DOE/Department-Of-Energy/Media-Center-Publications/1_Abu-Dhabi-Energy-Outlook-2050_En.pdf) /58/. The validation team noted that the report was prepared by The Abu Dhabi Department of

	Energy (DoE) and hence found the information, credible. From Page 17 of the report, the validation team noted that the penetration rate of district cooling technology is 11% in 2021. It is also stated in the same page that the penetration rate will remain 11% in absence of climate policies. In the third-party report, carried out by Global Market insights, penetration rate of the district cooling technologies is determined as 18%, which is more conservative and therefore has been used in the analysis. Thus the validation team ensured that the penetration rate of the district cooling technologies is determined as per the applied methodology.
19. The project activity is deemed to be additional if: (a) The seasonal energy efficiency ratio (SEER) of the district cooling plant involved in the project activity (determined as per section 5.2.3.2) is higher than a benchmark SEER (determined as per section 5.2.3.3); and (b) The share of the district cooling technologies at the moment of the project registration is less than 20 per cent of all cooling technologies within the benchmark boundary in terms of cooling output. The share of technologies can be derived from official country reports, third party	Applicable (a) The SEER (4.03) of Project is Higher than the benchmark SEER (2.26). The description on VVB assessment is provided in additionality section below. (b) Share of district cooling technologies in the UAE 18% which is less than 20% of total cooling output. A third-party survey report is submitted to validator /12/.

surveys and/or credible international sources (e.g. International Energy Agency).	
20. If any of two conditions above is not met, the project proponent should proceed to Step 2.	Not applicable Both the conditions above are meeting.

Validation team confirms that the baseline scenario opted by the project activity is in accordance with the requirements of the applied methodology/40/ and is justified.

Findings: CL 02 is raised and successfully closed. (Refer Appendix 3 for further details)

3.3.5 Additionality

The steps taken to validate that the procedure for additionality (set out in the methodology) has been followed correctly and precisely, are described below:

Benchmark determination procedure

The benchmark value serves for two purposes: (a) additionality demonstration and (b) baseline emissions determination.

Benchmark boundary

As per the applied Methodology the default benchmark boundary is a city where the project activity is located which is Abu Dhabi. However, the validation team noted that the methodology does not provide any definition related to city. The PP has submitted a methodology deviation for selection of benchmark boundary in section 3.6 of VCS PD /1.2/. The validation team noted that this deviation meets the requirements of the applicable VCS standard:

The validation team noted that the host country UAE is divided into seven administrative divisions² (Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah), referred as 'Emirates'. Each Emirates have its own rules and regulations and clear demarcation of boundary. The PP has considered Abu Dhabi Emirate as benchmark boundary for demonstrating methodological requirements. As this is a grouped project and therefore the boundary is limited to Abu Dhabi Emirate, United Arab Emirates. The validation team noted that the Emirates have a high demand for air conditioning. From section 2.3.1.2 of the baseline study report by Global Market Insights Inc /47/, the validation team noted that the Prevalent elevated temperatures build unfavorable living conditions, which make air conditioning a

² https://en.wikipedia.org/wiki/Politics_of_the_United_Arab_Emirates

necessity across the Emirates. Hence, the selection of benchmark boundary as Abu Dhabi Emirates is found to be reasonable. Hence this deviation is found to be acceptable.

The validation team noted that as per Para 36 of the applied methodology, the first benchmark value shall be identified before the start of the validation process for the purpose of additionality demonstration and ex-ante emission reduction calculations.

VCS-PD was published for Global Stakeholder comments in VERRA registry from 21-August-2023 to 20-September-2023, which is the start of the validation process.

The benchmark study report, prepared by Global Market insights, considered historical data for the period from 2019 to 2022 for the identification of Benchmark SEER value; thus satisfying the requirements of Para 36. Also, as per para 33 of the applied methodology, data for the most recent three years (2019 to 2022) is used to determine the values of SEER for the existing installations included in the benchmark boundary. Therefore, no new building (as per the definition of the applied methodology) was considered in the Benchmark study. As per the definition of 'New Buildings' Provided in Section 4 of the applied methodology 'New buildings (sub-area) - buildings within the project boundary that are constructed after the implementation of the project activity and are supplied district cooling by the project activity;'

The validation team noted that as per para 26 of the applied methodology, 'New buildings shall be included in the benchmark determination. For this purpose, new buildings are assumed to be existent before the start of the project and use the best available technology (BAT) in a host country'.

The validation team noted that the Benchmark study, carried out by Global Market insights, has included 'Greenfield Buildings' (implemented until 2022) for the determination of Benchmark SEER value. The study also determined that the best available technology in the host country is district cooling plant with SEER 4.84 (MWh/MWh). From section 2.5 of the report /12/, the validation team noted that the district cooling system is ranked 1, among the technologies considered for the determination of Benchmark SEER value. Thus, validation team noted that the Benchmark study carried out by Global Market insights satisfies the requirements of Para 26 of the applied methodology.

The PO stated that since the benchmark value is valid only for three years (as per para 36 of the applied methodology), in the next update of benchmark study report (after 3 years), new buildings (*building implemented after the implementation of the project activity*) will be included in the benchmark boundary.

Thus, the validation team ensured that the benchmark seasonal energy efficiency ratio (SEER) is determined in line with the methodology requirement.

Data vintage and data source

The Validation team noted that the Benchmark SEER has been determined using SEERs values and cooling output of the installations used within the benchmark boundary i.e. Abu Dhabi Emirate. This meets the requirement of para 27 of the applied methodology.

The project owner has appointed a third Party, Global Market Insights Inc to carry out the Abu Dhabi cooling sector analysis to provide the benchmark and penetration of cooling technologies /12/.

The validation team noted that in the benchmark boundary all the operational installations have been considered which can be classified as existing installation.

The third party has collected the data from various technology manufacturers, cooling providers, cooling technology users, government entities working in Abu Dhabi and provided independent report. The main data is presented in section below.

The values of SEER for the existing installations included in the benchmark boundary has been determined based on option 1 in para 29 of the applied methodology i.e. Directly from manufacturer of the baseline cooling technology as per the methodology.

The benchmark study report, prepared by Global Market insights, considered historical data for the period from 2019 to 2022 for the identification of Benchmark SEER value; thus satisfying the requirements of Para 36. Also, as per para 33 of the applied methodology, data for the most recent three years (2019 to 2022) is used to determine the values of SEER for the existing installations included in the benchmark boundary.

The SEER Values of all the installation arranged by type of technology is presented in the table below³: The validation team noted that this table is provided in the baseline study report.

Technology	SEER Rating (BTU/W-Hr)	SEER (MWh/MWh) ⁴
District Cooling System	16.5	4.84
Split Units	15.4	4.51
Centralized Water Cooled	14.7	4.31
Others (Ducted & Packaged Cooling Systems)	12	3.52
Air Cooled Stand Alone	7.7	2.26

Benchmark determination

The PP has applied para 35 of the applied methodology for Benchmark determination. The PP has submitted benchmark calculation sheet /60/, separately. The information presented in the

³ Page 56, section 2.6 Abu Dhabi cooling market report by GMI

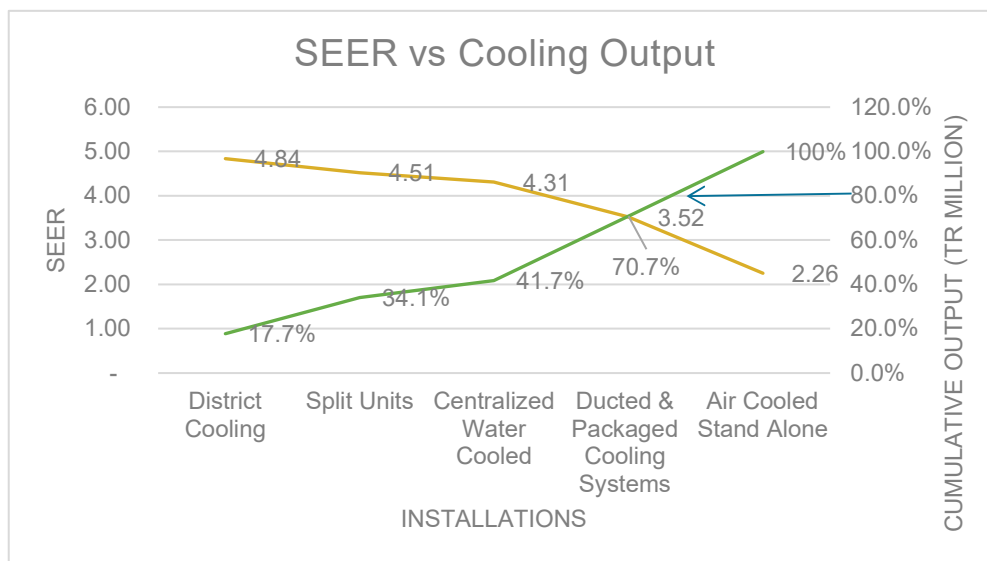
⁴ Conversion is divided BTU/Wh by 3.412 to convert in MWh/MWh

sheet are sourced from the third party report /12/. The following steps are applied in Benchmark determination:

(a) All installations included in the benchmark boundary are ranked in order of decreasing SEER; The PP has included a table in the VCS PD /1.2/, which is taken from Page 55 of the Baseline study report /47/. The table is reproduced below.

Technology	SEER	Cooling Output in million TR ⁵	Cumulative Cooling Output (million TR)
District Cooling	4.84	0.63	0.63
Split Units	4.51	0.58	1.21
Centralized Water Cooled	4.31	0.27	1.48
Others (Ducted & Packaged Cooling Systems)	3.52	1.03	2.51
Air Cooled Stand Alone	2.26	1.04	3.55
Total		3.55	3.55

b) The SEER values (y axis) as a function of the cumulative cooling output (x-axis) is plotted in a graph. The PP has included a graph in the VCS PD /1.2/, which is taken from Page 56 of the Baseline study report /47/. The graph is reproduced below.



c) The benchmark installation that corresponds to at least 80 per cent of the cumulative cooling output is identified from the graph. From the graph, it is noted that Benchmark installation at 80% is Air Cooled Stand alone cooling system.

⁵ Tons of refrigeration

d) The value of the benchmark SEER that corresponds to at least 80 per cent of the cumulative cooling output determined in step (c) above is matched. It is noted that the SEER value for that installation is 2.26 MWh/MWh.

Validity of the benchmark

The first benchmark value has been identified before the start of this validation process for the purpose of additionality demonstration and ex-ante emission reduction calculations. As per the requirements of Para 36 of the applied methodology, the benchmark value for baseline emissions estimation is valid for three years. Based on interview with the PP, the project proponent informed that the benchmark will be updated every 3 years after registration of the project, as per the requirements of para 36 of applied methodology. The validation team has raised FAR 02 in this regard.

The PP has submitted the benchmarking report /47/ during validation.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

The equation and choices provided in the methodology and all other methodological tools are correctly quoted in the PD /1.2/. The emission reductions of the project instances of the grouped project would be calculated using the formulae mentioned in the applied methodology; AM0117 (version 2.0)/40/.

Validation team based on the review of the PD /1.2/, confirms that the formulae are correctly presented for the determination of emissions reductions at project instance level. The parameters and equations presented in the PD/1.2/, as well as other applicable documents, have been compared with the information and requirements presented in the methodology respectively. An equation comparison has also been made to ensure consistency between all the formulae presented in the PD/1.2/ and ER spreadsheet/2.2/ and methodology AM0117 (version 2.0)/40/.

According to applied methodology AM0117 (version 2.0) /40/ the emissions are calculated as below:

Baseline emissions:

The validation team noted that the PP has used Approach 1 for estimating Baseline emissions as the benchmark installation is use of electricity driven technology.

Total baseline emissions are calculated as follows, using Equation (2) from the methodological tool 05; Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation version 3.0. When applying the tool, parameter QB,y shall correspond to parameter EC_{BL,k,y}, whereas BE_y shall correspond to BE_{EC,y}.

$$BE_y = Q_{B,y} \times EF_{B,y} \times (1 + TDL_{B,y})$$

Equation
2 from
Tool 05

Where

Parameter	Description	Value used for estimation	VVB Assessment
BE _y	Baseline emissions in year y (tCO ₂ e/yr)	36,393	Calculated using Equation (2) from the methodological tool 05; Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation version 3.0.
Q _{B,y}	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year y (MWh/yr)	73,614 MWh/year	Calculated, based on Equation 7 of the applied methodology. The calculation is explained further, below.
TDL _{BY}	Average technical transmission and distribution losses for providing electricity to the baseline in year y	8.18%	Department of energy '2021 Annual Technical Report) /25/. Page 31 for transmission losses (2.18%) and page 41 for distribution losses (6%)
EF _{EL,y}	Grid CO ₂ emission factor calculated in accordance with the "Tool to calculate the emission factor for an electricity system" (Version 7.0) (tCO ₂ e/MWh)	0.457	Publicly available grid emission factor from International Renewable Energy Agency, for the host country -. IRENA UAE Energy profile published on 8th August 2023 /23/.

The estimate cooling output of isolated and less efficiency cooling technologies is determined as follows:

$$Q_{B,y} = \sum_r \frac{OPC_{r,y}}{SEER_{B,i}} \quad \text{Equation (1)}$$

Where:

Parameter	Description	Value used for estimation	VVB Assessment
$Q_{B,y}$	Energy consumed in baseline by baseline cooling technologies in year y (MWh)	73,614 MWh/year	Calculated, based on Equation 7 of the applied methodology.
$OPC_{r,y}$	Cooling output of new district cooling plant r in year y (MWh)	166,368	Estimated based on project capacity (9000 Tons of refrigeration), 8760 hours/year, 60% estimated annual load factor ⁶ , 3.517 is for conversion from TRh to kWh and division by 1000 to convert to MWh.
$SEER_{B,i}$	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i, defined as energy output divided by energy input (MWh/MWh)	2.26	As per the methodology procedure based on data provided by Third party. /12/

The grid CO₂ emission factor is calculated in accordance with the "Tool to calculate the emission factor for an electricity system" (Version 07) for the respective grid ex-ante and is fixed throughout the crediting period.

The following steps are applied:

STEP 1 Identify the relevant electricity systems;

STEP 2 Choose whether to include off-grid power plants in the project electricity system (optional);

STEP 3 Select a method to determine the operating margin (OM);

STEP 4 Calculate the operating margin emission factor according to the selected method;

STEP 5 Calculate the build margin (BM) emission factor;

⁶ Based on past operations of other district colling plant, Project is expected to have 60% annual load factor. This is based on project proponents internal assessment.

STEP 6 Calculate the combined margin (CM) emission factor.

The cooling output of the new District Cooling plant, $OPC_{r,y}$ is determined in MWh, as per below equation. The PP uses Option 1: Direct measurement of the cooling energy, provided in the applied methodology. Hence the selection of Equation 9 of the methodology is correct.

$$OPC_{r,y} = c_p \times F_{p,r,y} \times \Delta T_{r,y} \times h_{r,y} \times 2.77 \times 10^{-10} \quad \text{Equation (2)}$$

Where:

$OPC_{r,y}$	Cooling output of new district cooling plant r in year y (MWh)
$F_{p,r,y}$	Average mass flow rate (integrated over the year) of new district cooling plant r in year y (g/hour)
$\Delta T_{r,y}$	Temperature difference between supply and return of chilled water from/to new district cooling plant r in year y (°C)
$h_{r,y}$	Number of the operating hours of the new district cooling plant r in year y (hours)
c_p	Specific heat capacity of coolant (J/g. °C)

Project Emissions

Project emissions include emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant.

These emissions are calculated using the “Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” version 3.0.

Equation 11 of
applied
methodology

$$PE_y = PE_{EC,y} + PE_{FC,j,y}$$

Where:

Parameter	Description	Value used for estimation	VVB Assessment
PE_y	Project emissions in year y (tCO ₂ e)	16,994	Calculated using Equation 11 of applied methodology. The PP has provided calculations in the ER sheet. The values used for this calculation are validated separately and explained below.
$PE_{EC,y}$	Emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant(s) (tCO ₂ e)	34,374	Calculated using Equation 2 of Tool 05. The PP has provided calculations in the ER sheet. The values used for this calculation are validated separately and explained below.
$PE_{FC,j,y}$	CO ₂ emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO ₂ e)	0	The project consumes electricity from grid. The electricity is purchased from the grid. The validation team confirmed the same from verifying the grid connection and through receipts of electricity purchases /53/. There is no fossil fuel captive power plant at site. Hence, this is not applicable.

Equation 2 of
Tool 05

$$PE_{EC,j,y} = EC_{PJ,j,y} \times EF_{EL,y} (1 + TDL_y)$$

Where

Parameter	Description	Value used for estimation	VVB Assessment

EC _{PJ,j,y}	Quantity of electricity consumed by the new district cooling plant j in year y (MWh/yr)	34,374	Estimated based on the cooling load from the project and SEER of district cooling plant (4.84 MWh/MWh). The validation team noted that during monitoring actual electricity consumption will be measured.
EF _{EL,y}	Grid CO ₂ emission factor calculated in accordance with the “Tool to calculate the emission factor for an electricity system” (Version 7.0) (tCO ₂ e/MWh)	0.457	Publicly available grid emission factor from International Renewable Energy Agency, for the host country -. IRENA UAE Energy profile published on 8th August 2023 /23/.
TDL _y	Average technical transmission and distribution losses for providing electricity in year y	8.18%	Department of energy '2021 Annual Technical Report) /25/. Page 31 for transmission losses (2.18%) and page 41 for distribution losses (6%)

Leakage

Leakage emissions are calculated as follows:

Equation 12 of

$$LE_y = LE_{Ref,y} + LE_{Ref,sc} + LE_{water}$$

Applied

methodology

Where:

Parameter	Description	Value used for estimation	VVB Assessment
LE_y	Leakage emissions in year y (tCO _{2e})	95 – first year 76- for subsequent years	Calculated using Equation 12 of applied methodology. The PP has provided calculations in the ER sheet. The values used for this calculation are validated separately and explained below.
$LE_{Ref,y}$	Refrigerant leakage emissions from the project in year y (tCO _{2e})	76	Calculated using Equation 13 of applied methodology. The PP has provided calculations in the ER sheet. The values used for this calculation are validated separately and explained below.
$LE_{Ref,sc}$	Refrigerant leakage emissions from the baseline cooling equipment that is scrapped as a result of the project activity (only accounted in the first year of the crediting period) (tCO _{2e})	0	The validation team noted that this is not applicable for project as no baseline equipments are scrapped. The project activity employs new district cooling plants.
$LE_{water,y}$	Emissions due to the freshwater usage in the project system in year y (tCO _{2e})	19	Calculated using Equation 15 of applied methodology. The PP has provided calculations in the ER sheet. The values used for this calculation are validated separately and explained below.

Emissions due to the refrigerants leakage during the project activity is calculated as:

$$LE_{Ref,y} = \sum_k R_{k,y} \times GWP_k$$

Equation 13 of
applied
methodology

Where:

Parameter	Description	Value used for estimation	VVB Assessment
$LE_{Ref,y}$	Refrigerant leakage emissions from the project in year y (tCO ₂)	76	Calculated
$R_{k,y}$	Quantity of refrigerant k filled in the district cooling system year y (t)	11.68 Tons	The value is from manufacturer's chiller data sheet. The PP has submitted the chiller data sheets - Product Data Sheet Chiller CSD /51a/ and Product Data Sheet Chiller VSD /51b/ where CSD is constant speed drive and VSD variable speed drive. Since the values are taken from chiller manufacturer, the validation team noted that the value is acceptable.
GWP_k	Global Warming Potential of the refrigerant k	1300	The value is sourced from IPCC 5 th Assessment report, which is the recent data, at the time of validation /33b/. The value is correctly taken for the type of refrigerant used. R 134 A. The type of refrigerant used is confirmed during the on-site inspection directly from the name plate details of the chillers /52/. (https://ghgprotocol.org/sites/default/files/ghgp/Global-Warming-Potential-Values%20%28Feb%2016%202016%29_1.pdf)

Emissions due to use of freshwater shall be considered in sites where freshwater is produced through the desalination. As emissions associated with the production of make-up water are

considered negligible, this source of leakage shall be considered once, i.e. at the first monitoring and calculated as below:

$$LE_{water} = Q_{water} \times EF_{water}$$

Equation 15
of the
applied
methodology

Where:

Parameter	Description	Value used for estimation	VVB Assessment
LE_{water}	Emissions due to the freshwater usage in the project system (tCO ₂ e)	19	Calculated using Equation 15 of applied methodology. The PP has provided calculations in the ER sheet. The values used for this calculation are validated separately and explained below.
Q_{water}	Maximum designed quantity of freshwater to be used in the project system (m ³)	1620.5	The validation report noted that the value is mentioned in page 28 in the Concept Design Report /14/. Since the report is prepared by a competent third party, specific to the design of the project activity, the validation team found the value to be acceptable.
EF_{water}	Emission factor associated with the production of freshwater (tCO ₂ e/t)	11.5 kg CO ₂ /m ³ (0.0115 ton CO ₂ /ton)	The data is based on publicly available information, Taqa sustainability report 2022, page 80 /49/. Since this value is sourced from a publically available source, the validation team found it acceptable. However, for the first monitoring period, the PP stated that data provided by supplier, Emirates water and electricity company, will be used.

Emission Reduction

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 1	36,393	16,994	95	19,304
Year 2	36,393	16,994	76	19,323
Year 3	36,393	16,994	76	19,323
Year 4	36,393	16,994	76	19,323
Year 5	36,393	16,994	76	19,323
Year 6	36,393	16,994	76	19,323
Year 7	36,393	16,994	76	19,323
Total	254,751	118,958	551	135,242

The total emission reduction is 135,242 tCO₂e over 7 years renewable crediting period.

Through the validation process KBS has confirmed that:

- All the assumptions and data used by the project participants are listed in the PD including their references and sources;
- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the PD;
- All values used in the PD are considered reasonable in the context of the VCS project activity;
- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, leakage and emission reductions;
- All estimates of the baseline emissions can be replicated using the data and parameter values provided in the PD.

Findings: CL 03 and CL 04 are raised and successfully closed. (Refer Appendix 3 for further details)

3.3.7 Methodology Deviations

The validation team noted that for the validation of this grouped project, the PP has submitted two methodology deviations. The validation team noted that this deviation meets the requirements of the applicable VCS standard and the assessment details are provided below:

Deviation 1. Grid emission factor:

The methodology refers to CDM Tool 07 for determination of electricity grid emission factor. However, the PP informed that data for calculation of OM, BM and CM are not available in public domain and therefore project proponent is not able to calculate this factor.

The project Proponent has requested to use the publicly available grid emission factor for the project, which will be fixed for first crediting period. For future crediting period, project proponent anticipates that the data will be available to calculate the grid emission factor and then the Tool 7 or any other Verra guidance will be used for the project.

Based on publicly available information, the following data is available for UAE grid emission factor:

Source	Latest year for which emission factor available	Value
Third update for second NDC of the UAE – 2023 page 17 ⁷	2019	0.55 tCO ₂ /MWh
Energy profile, United Arab Emirates published by IRENA – 8 th August 2023 page 3 ⁸	2021	457 tCO ₂ /GWh (0.457 tCO ₂ /MWh)

The validation team noted that the value is sourced from Energy profile for United Arab Emirates published by International Renewable Energy Agency /23/. This is the latest (released in 2021), available at the time of validation. The value is also found to be lesser than the value available in the Third update for Second NDC of the UAE -2023. This value is also used for Baseline emission calculation. Hence the net effect of this value (*Baseline Emission calculation vs Project Emission calculation*) in ER calculation is conservative. Therefore, this has been used for the first crediting period. The deviation is found to be acceptable.

Deviation 2. Benchmarking Boundary

As per the applied Methodology, the default benchmark boundary is a city where the project activity is located. However, the validation team noted that the methodology does not provide any definition related to city.

⁷ https://unfccc.int/sites/default/files/NDC/2023-07/Third%20Update%20of%20Second%20NDC%20for%20the%20UAE_v15.pdf (page 17)

⁸ https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Middle-East/United-Arab-Emirates_Middle-East_RE_SP.pdf (page 3)

The validation team noted that the host country UAE is divided into seven administrative divisions⁹ (Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah), referred as 'Emirates'. Each Emirates have its own rules and regulations and clear demarcation of boundary. The PP has considered Abu Dhabi Emirate as benchmark boundary for demonstrating methodological requirements. As this is a grouped project and therefore the boundary is limited to Abu Dhabi Emirate, United Arab Emirates. The validation team noted that the emirates have a high demand for air conditioning. From section 2.3.1.2 of the baseline study report by Global Market Insights Inc /47/, the validation team noted that the Prevalent elevated temperatures build unfavorable living conditions, which make air conditioning a necessity across the Emirates. Hence, the selection of benchmark boundary as Abu Dhabi Emirates is found to be reasonable. Hence this deviation is found to be acceptable.

3.3.8 Monitoring Plan

The monitoring plan is in line with the applied methodology AM0117, version 2.0 /40/. The monitoring plan will give opportunity for real measurement of emission reductions achieved. The action plan for monitoring has been included in section 5.3 of the VCS PD /1.2/.

- Data and parameters available at validation

Following parameters are fixed for the entire crediting period and are mentioned in section 5.1 of the VCS PD/1.2/

Parameter	Unit	Value			Source/Justification
Building Type	NA	NA			The project will be providing the cooling demand to only new buildings. This is confirmed through interview with the PP. Hence, the building types is categorized as new.
Baseline cooling technologies	-	Technology	SEER	Cooling Output in million TR	The validation team noted that data collection on baseline cooling technologies has been done by third party, Global Market Insights
		District Cooling	4.84	0.63	
		Split Units	4.51	0.58	

⁹ https://en.wikipedia.org/wiki/Politics_of_the_United_Arab_Emirates

		<table> <tr> <td>Centralized Water Cooled</td> <td>4.31</td> <td>0.27</td> </tr> <tr> <td>Others (Ducted & Packaged Cooling Systems)</td> <td>3.52</td> <td>1.03</td> </tr> <tr> <td>Air Cooled Stand Alone</td> <td>2.26</td> <td>1.04</td> </tr> <tr> <td>Total</td> <td></td> <td>3.55</td> </tr> </table>	Centralized Water Cooled	4.31	0.27	Others (Ducted & Packaged Cooling Systems)	3.52	1.03	Air Cooled Stand Alone	2.26	1.04	Total		3.55	/12/ based on various data sources, including Abu Dhabi Department of Energy (DoE), Abu Dhabi Industrial Strategy (ADIS), Abu Dhabi Department of Economic Development (ADDED), Abu Dhabi Industrial Development Bureau (IDB). Hence, the values determined are found acceptable.
Centralized Water Cooled	4.31	0.27													
Others (Ducted & Packaged Cooling Systems)	3.52	1.03													
Air Cooled Stand Alone	2.26	1.04													
Total		3.55													
EF _{grid,CM,y} Combined margin emission factor for the grid	t CO ₂ /MWh	0.457	Due to unavailability of the data for calculation of OM, BM and CM, project proponent has requested for a deviation to use publicly available grid emission factor from International Renewable Energy Agency, for the host country -. IRENA UAE Energy profile published on 8 th August 2023 /23/. The assessment of the deviation is described in detail in section 3.3.7 of this report.												

			Hence, the value determined is found acceptable.
Q _{water} , Maximum designed quantity of freshwater to be used in the project system	Ton	1620.52	The validation team noted that the value is mentioned in page 28 in the Concept Design Report /14/. Since the report is prepared by a competent third party, specific to the design of the project activity, the validation team found the value to be acceptable.

Findings: CAR 02 is raised and successfully closed. (Refer Appendix 3 for further details).

- Data and parameters monitored

Following parameters are monitored during the crediting period and are mentioned in section 5.2 of the VCS PD /1.2/:

Parameter	Unit	Source/Justification
OPC,r,y Cooling output of new district cooling plant r in year y	TRh	The PP has stated that this parameter will be monitored through online monitoring system at the site. The PP has provided frequency of monitoring to be continuous – based on on-site metering systems. PP has also provided a cross checking procedure – to be cross checked with purchase orders. QA/QC procedures are defined – Calibration of scales used for measuring the weight of cylinders. The procedures described for monitoring and measurement arrangements of the parameter are deemed to be feasible, meeting the requirements of the methodology and hence accepted.

$EC_{PJ,j,y}$ Quantity of electricity consumed by the new district cooling plant j in year y	<i>MWh</i>	The PP has stated that this parameter will be monitored through online monitoring system at the site. The PP has provided frequency of monitoring to be continuous – based on on-site metering systems. PP has also provided a cross checking procedure – to be cross checked with purchase orders. QA/QC procedures are defined – Calibration of scales used for measuring the weight of cylinders. The procedures described for monitoring and measurement arrangements of the parameter are deemed to be feasible, meeting the requirements of the methodology and hence accepted.
$TDL_{B,y}$ and $TDLy$ Average technical transmission and distribution losses for providing electricity in year y	%	The PP has stated that the value for this parameter will be Annual average value based on the most recent data available within the host country. The procedures described for monitoring and measurement arrangements of the parameter are deemed to be feasible, meeting the requirements of the methodology and hence accepted.
$SEER_{B,i}$ Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i	MWh/MWh	The source of data for this parameter are third party report (which are based on Records from manufacturers of the baseline cooling technology). The PP has provided frequency of monitoring to be once in 3 years for included project instance. as and when the cylinders are used. The procedures described for monitoring and measurement arrangements of the parameter are deemed to be feasible, meeting the requirements of the methodology and hence accepted.
$R_{k,y}$ Quantity of refrigerant k used in the project in year y	T	The source of data for this parameter will be Records from the plant operator on the simple counting of cylinders /32/. The PP has provided frequency of monitoring to be continuous – as and when the cylinders are used. PP has also provided a cross checking procedure – to be cross checked with purchase orders. QA/QC procedures are defined – Calibration of scales used for measuring the weight of cylinders. The procedures described for monitoring and measurement arrangements of the

		parameter are deemed to be feasible, meeting the requirements of the methodology and hence accepted.
GWPK, Global Warming Potential of the refrigerant k	--	The validation team noted that the PP has taken the value Global Warming Potential Values from fifth assessment report /33b/. The validation team has raised a FAR 03 to update the value for future commitment periods according to any future COP/MOP.
EF _{water} Emission factor associated with the production of fresh water	tCO _{2e} /t	This data is from official source - Emirates water and electricity company. The value will be the latest available at the time of first monitoring period.

Validation team confirms that the overall monitoring plan complies with the requirements of the methodology AMO117 /40/; the monitoring arrangements described in the monitoring plan are feasible within the project design and the project proponents will be able to implement the described monitoring plan.

Findings: CAR 03 is raised and successfully closed. (Refer Appendix 3 for further details)

3.4 Non-Permanence Risk Analysis

Non-permanence risk analysis is not applicable as the Project is not an AFOLU (Agriculture, Forestry and Other Land Use) project.

4 VALIDATION OPINION

4.1 Validation Summary

National Central Cooling Company PJSC - Tabreed has commissioned KBS Certification Services Limited. to do VCS validation of the project activity: “**Tabreed District Cooling Group Project**” having VCS ID 4519, with regard to the relevant requirements of VCS Standard version 4.5; to attain real, measurable, additional and permanent emission reductions.

The project involves installing new district cooling plants of various capacities to supply cooling to residential and commercial consumers in Abu Dhabi, United Arab Emirates. The Project will replace isolated and less efficient cooling technologies (such as air-cooled reciprocating chiller systems) and thereby reducing the consumption of electricity and corresponding Carbon Dioxide (CO₂) emissions.

A risk-based approach has been followed to perform this validation. In the course of the validation 03 Corrective Action Request (CAR) and 04 Clarification Request (CL) were raised and successfully closed. 03 FARs raised during the validation.

The validation is based on the VCS PD, additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and supporting documents made available to the validation team by project proponent.

As a result of the validation, the validation team confirms that:

- The GHG statement is the responsibility of the PP.
- The project confirms with the validation criteria for the projects set out in VCS Standard Version 4.5.
- The project is in line with all relevant VCS requirements.
- The project baseline is sufficiently justified in the PD.
- The calculation of the project emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of 135,342 tCO₂e is most likely to be achieved within the 7 years of renewable crediting period for the first project instance.
- No restrictions or uncertainties were identified related to the validation. There is no adverse, disclaimed, modified or qualified opinion in this validation.
- The validation of the GHG statement was conducted in accordance with ISO 14064-3:2019)

4.2 Validation Conclusion

The data and information supporting the GHG statement assertion were projected. The validation report is based on VCS-PD, supporting documents made available to the validation team and information collected through performing interviews. Based on the process and procedures conducted, KBS concludes that the information in the PD:

- is materially correct and is a fair representation of the actual project details, and
- is prepared in accordance with VCS requirements and the applied methodology for information pertaining to description, methodological deviations and reporting.

Through the on-site interview, host country rule and regulations related to project activity, Project description, technological measures, Implementation, Operation, Management of project activity and Training of personnel, Stakeholder consultation etc. has been checked and found appropriate. The validation work is carried out as per this requirement and the validation opinion is assured provided the credibility of all above. The actual results may vary since the estimates are based on assumptions that are subject to change. The project is likely to achieve the estimated GHG emission reductions and carbon dioxide removals described below.

Crediting period: From 25-May-2023 to 24-May-2030

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

The project falls under the category for which it is not required to assess permanence risk.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
25-May-2023 to 31-December-2023	22,035	10,290	65	11,681	Not applicable	11,681
01-January-2024 to 31-December-2024	36,393	16,994	76	19,323		19,323
01-January-2025 to 31-December-2025	36,393	16,994	76	19,323		19,323
01-January-2026 to 31-December-2026	36,393	16,994	76	19,323		19,323
01-January-2027 to 31-December-2027	36,393	16,994	76	19,323		19,323
01-January-2028 to 31-December-2028	36,393	16,994	76	19,323		19,323

01-January-2029 to 31-December-2029	36,393	16,994	76	19,323		19,323
01-January-2030 to 24-May-2030	14,358	6,704	30	7,623		7,623
Total	254,751	118,958	551	135,242		135,242

For projects required to assess permanence risk: Not applicable.

Location: Faridabad, Haryana, India

Date: 30-May-2024



Kaushal Goyal

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

There are no commercially sensitive information that the project proponent would like to exclude in the public version.

APPENDIX 2: REFERENCES

Sr. No.	Name of Document
01	/1.1/ - Webhosted VCS PD, Version 01, Dated 21-June-2023 /1.2/ - Final VCS PD, Version 1.9, Dated 10-May-2024
02	/2.1/ - Emission reduction calculation sheet (Submitted with VCS PD, Version 1) – with File Name - “Emission Reduction calculation-20230713” /2.2/ - Emission reduction calculation sheet (Submitted with VCS PD, Version 1.9) – with File Name - “ Emission Reduction calculation_05 Feb 2024”
03	Supporting evidence stating avoidance of double counting in any other GHG emission reduction mechanism – Declaration issued by TABREED dated 03-Sep-2023
04	Technical Specification documents for the key project equipments - Product data sheets for chillers issued by Johnson Controls dated 24-Jan-2021.
05	Commissioning certificate issued by Main Equipment supplier (Chiller)- Johnson Controls and accepted by PP, TABREED: Evidence to support the operation start date: 25-May-2023. - Supporting evidence for project start date
06	Contract between PP and End user on ownership of the generated emission reductions.
08	Musataha Agreement between Miral Asset Management LLC and National Central Cooling Company PJSC, in respect of a District cooling plant site at the SeaWorld Abu Dhabi Development on Yas Island, Abu Dhabi
09	a. District cooling technical code, Issued by Department of Energy, effective date 15-March-2020. b. District cooling Metering code, issued by Department of Energy, effective date 15-March-2020.
10	Equipment purchase order - Chiller- issued by TABREED to Johnson Controls Air Conditioning & Refrigeration Inc, dated 16-11-2021
11	Approvals/Clearances from Government authorities a. Building Permit from Department of Municipalities and Transport dated 06/01/2022
12	Benchmarking report (for the purpose of additionality demonstration and ex-ante emission reduction calculations) – Prepared by Global Market insights dated September, 2023.
13	Maps or schematic plan diagrams of the district cooling system and the area where project is implemented obtained from the National District cooling company PJSC , TABREED.
14	District cooling plant design documentation, prepared by Fittons Engineering Consultancy LLC ‘ P131-RP-M-08 R00 dated 15-January-2021.
15	Document (KML file) supported with GPS coordinates.

16	Technical report/feasibility study of 3rd party on the penetration of district cooling in Abu Dhabi, Global Market Insight Sep 2023
18	Technical Data of chiller plant and Thermal Energy storage systems.
19	Declaration of no ODA - Letter issued by TABREED dated 03-Sep-2023
20	Declaration of no participation in other GHG programs - Letter issued by TABREED dated 03-Sep-2023
21	Environment permit from Environmental Agency – Abu Dhabi (Licence No. CE-237) dated 09/03/2023 to the proponent, National Center Cooling Company PJSC for construction and operation.
22	Environmental Technical Questionnaire/Application submitted to Environmental Agency- Abu Dhabi
23	Evidence for Grid Emission factor for UAE – UAE Energy profile document published by IRENA https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Middle-East/United-Arab-Emirates_Middle-East_RE_SP.pdf
24	Third update for second NDC of the UAE – 2023 page 17 https://unfccc.int/sites/default/files/NDC/2023-07/Third%20Update%20of%20Second%20NDC%20for%20the%20UAE_v15.pdf
25	2021 Annual Technical Report from Department of Energy
26	Licence Application submitted to Department of Energy for the project activity
27	Email communication dated 24-May-2023 from Department of Energy, addressing to the PP, TABREED, confirming the Design capacity and Installed capacity.
28	Evidence for Education and training of staff in the operation of the district cooling system – Employee training records from May 2023 to August 2023
29	Model beneficiary agreement with end users – Sample agreement submitted by PP, TABREED.
30	Local stakeholder consultation: a. Media release about the project activity, issued on January 2021 b. Invitation to the meeting, conducted in September 2023 c. Photographs, Minutes of the meeting, Feedback received from stakeholders and their responses for the meeting conducted in September 2023.
31	Evidence for purchase of electricity from grid for the project activity.
32	Evidence for Type and Quantity of refrigerant (<i>Used in the project activity</i>) – Log books Records from the plant operator on the simple counting of cylinders

33a	Evidence for Type and Quantity of refrigerant (<i>Used in Baseline individual cooling Equipment</i>) – <i>Baseline study report issued by Global Market Insights</i>
33b	Evidence for Global warming potential of refrigerant – IPCC 5 th Assessment report - https://ghgprotocol.org/sites/default/files/ghgp/Global-Warming-Potential-Values%20%28Feb%2016%202016%29_1.pdf
34	Evidence for source and quantity of freshwater for the project activity- Log book records verified during the site visit for the period from June 2023 to August 2023
35	Baseline Cooling Technology (for the last 3 years) -Number and type of buildings supported by the technology in the project boundary -Annual Cooling output from the baseline technology
36	Concept Design Report for Sea World District Cooling plant, Aby Dhabi, prepared by Fittons Engineering Consultancy LLC dated 15/01/2021
37	Purchase order issued to Johnson Controls Air Conditioning & Refrigeration INC for supply of 3 nos x 3000 TR Chillers dated 16/11/2021.
38	Actual plant operation daily data for the period June 2023 to August 2023- submitted as '2-3 Seaworld DC plant – daily cooling and power consumption data'
39	Operation process flow diagram provided in Commercial facilities application
40	Applied Methodology UNFCCC CDM Methodology AM0117: Introduction of a new district cooling system; Version 02
41	Applied tools a. Tool for the demonstration and assessment of additionality (version 07) b. Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion (version 03) c. Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03) d. Tool to calculate the emission factor for an electricity system (version 07) e. Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period (version 03.0.1) f. Additionality of first-of-its-kind project activities (version 03.0)
42	CDM Requirements a. Glossary of CDM Terms, Version 11.0 b. General guidelines for SSC CDM methodologies, Version 23.1 c. General guidance on leakage in biomass project activities, version 03

43	VCS Requirements a. VCS Standard, version 4.5 Updated 04-October-2023 b. VCS Program Guide, version 4.4 Updated 29-August-2023 c. VCS Validation and Verification Manual, version 3.2 Updated 19-October-2016 d. Registration & Issuance Process, version 4.4 Updated 04-October-2023 e. VCS Program Definitions version, version 4.4 Updated 29-August-2023 f. VCS Monitoring Report Template version 4.3, Updated 29-August-2023 g. VCS Project Description Template, version 4.3, Updated 29-August-2023
44	Observations from On-site visit and interview, conducted on 25-September-2023.
45	VERRA review comments during pipeline listing of PD
46	Host country regulations
47	Baseline study report by Global Market Insights Inc dated September 2023
48	Electricity statistics provided by Abu Dhabi Electricity and Water Authority - 2021
49	Taqva sustainability report 2022
50	Chiller refrigerants: Minimizing carbon footprint, Johnson control (Chiller manufacture)
51	a. Product data sheet Chiller, CSD – issued by Key equipment supplier, Johnson controls dated 24-Jan-2021 b. Product data sheet Chiller, VSD - issued by Key equipment supplier, Johnson controls dated 24-Jan-2021
52	Name plate details of chillers – Photographs taken by KBS assessment team during the on-site visit.
53	Receipts of electricity purchases for the period – June 2023 to August 2023
54	Communication agreement with VERRA
55	Notice of Validation/Verification Services(NOVs) form, sent on 05-Sep-2023 to 'Auditing' <auditing@verra.org>
56	https://www.tabreed.ae/wp-content/uploads/2023/07/Tabreed ESG Report 2022.pdf
57	Validation contract with KBS Certification dated 03-June-2023.
58	Abu Dhabi Energy Outlook 2050
59	Diagram for Cooling demand Distribution network from DCP to Sea world
60	Benchmark Calculation sheet
61	Newspaper article for Inauguration of Sea world facility on 23-May-2024

APPENDIX 3: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Personnel Name		Srinivasan Selvaraj			
Schemes	☒ CDM	☒ GCC	☒ GS	☒ VCS	☐ Other GHG Schemes (mention here)
Qualified to work as					
Team Leader			☒	Technical Expert	☒
Validator/Verifier			☒	Financial Expert	☐
Technical Reviewer			☐	Local Expert (India)	☒
Area(s) of Technical Expertise					
Sectoral Scope			Technical Area		
SS 01: Energy Industries (Renewable/non-renewable sources)			TA 1.2: Energy generation from renewable energysources		
SS 03: Energy Demand			TA 3.1 : Energy Demand		
SS 5: Chemical industry			TA 5.1 Chemical industry		
SS 12: Solvents use			TA 12.1 Chemical industry		
SS 13: Waste handling and disposal			TA 13.1. Waste handling and disposal		
Approved by (Manager Competence & Training)			Dr. Rajesh Monga		
Approval date			25-08-2023		

Personnel Name		S Thyagaraj			
Schemes	☒ CDM	☒ GCC	☒ GS	☒ VCS	☐ Other GHG Schemes (mention here)
Qualified to work as					
Team Leader			☒	Technical Expert	☒
Validator/Verifier			☒	Financial Expert	☒
Technical Reviewer			☐	Local Expert	☒
Area(s) of Technical Expertise					
Sectoral Scope			Technical Area		
Energy industries (renewable/non-renewable sources)			TA 1.2: Energy generation from renewable energy sources		

Energy industries (renewable/non-renewable sources)	TA 1.1: Thermal energy generation from fossil fuels and biomass including thermal electricity from solar
Manufacturing industries	TA 4.1: Cement and lime production
Chemical industry	TA 5.1 Chemical industry
Metal production	TA 9.2: Iron, Steel, and Ferro-alloy production
Solvents use	TA 12.1: Chemical industry
Waste Handling and Disposal	TA 13.1: Waste Handling and Disposal
Approved by (Manager Competence & Training)	Manager C&T
Approval date	16-03-2022

Personnel Name		Sauvik Banerjee			
Schemes	☒ CDM	☒ GCC	☒ GS	☒ VCS	☐ Other GHG Schemes (mention here)
Qualified to work as					
Team Leader			☒	Technical Expert	☒
Validator/Verifier			☒	Financial Expert	☐
Technical Reviewer			☐	Local Expert	☒
Area(s) of Technical Expertise					
Sectoral Scope			Technical Area		
SS 1: Energy industries (renewable/non-renewable sources)			TA 1.2: Energy generation from renewable energy sources		
SS 1: Energy industries (renewable/non-renewable sources)			TA 1.1: Thermal energy generation from fossil fuels and biomass including thermal electricity from solar		
SS 3: Energy Demand			TA 3.1 Energy Demand		
Approved by (Manager Competence & Training)			Manager C&T		
Approval date			18-04-2022		

Personnel Name		Khushboo Singh			
Schemes	☒ CDM	☒ GCC	☒ GS	☒ VCS	☐ Other GHG Schemes (mention here)
Qualified to work as					
Team Leader			☐	Technical Expert	☐
Validator/Verifier (Trainee)			☒	Financial Expert	☐
Technical Reviewer			☐	Local Expert	☐
Area(s) of Technical Expertise					
Sectoral Scope			Technical Area		
-			-		
Approved by (Manager Competence & Training)			Shikha Sharma		
Approval date			11-01-2023		

Personnel Name:		S. Ranganathan	
Qualified to work as:			
Team Leader	☒	Technical Expert	☒
Validator/Verifier	☒	Financial Expert	☐
Technical Reviewer	☒	Local Expert (India)	☒
Area(s) of Technical Expertise			
Sectoral Scope	Technical Area		
SS 01: Energy industries (renewable/non-renewable sources)	TA 1.1: Thermal energy generation from fossil fuels and biomass including thermal electricity from solar		
	TA 1.2: Energy generation from renewable energy sources		
SS 2: Energy distribution	TA 2.1. Energy distribution		
SS 3: Energy demand	TA 3.1. Energy Demand		
SS 5: Chemical industry	TA 5.1 Chemical industry		
SS 12: Solvents use	TA 12.1 Chemical industry		
SS 13: Waste handling and disposal	TA 13.1 Waste Handling and Disposal		
	TA 13.2 Manure		
Approved by (Manager C& T)	Shikha Sharma		
Approval date:	05/05/2022		

APPENDIX 4: FINDINGS

Project Title:	Tabreed District Cooling Group Project
CDM/GS/GCC/VCS Ref No:	VCS Ref no:- 4519
MONITORING PERIOD (IF APPLICABLE):	-

General Instructions:

- The findings raised shall be clear manner giving reference to the criteria against which it was compared, the observation by the audit team, the spotted deviation and suitable action e.g. clarification/justification/correction etc., as appropriate.
- The findings shall not be removed at any stage by any audit team. However, in a rare situation if a finding is removed it can only be done by TR e.g. two identical findings are raised erroneously. Under these situations, the removal shall be documented in CDM-D-35.
- The response from PP shall be clear and precise to the nature of the findings raised and should contain information where the change has been made e.g. which section of the PDD.
- The resolution shall be specific to the point raised and reported in a clear manner by the audit team referencing the underlying sources of information used.
- The findings can be added by the audit team during the validation based on new insight or new requirements that comes in place.

- The findings raised by technical reviewer may also be raised here, as appropriate.
- The comments raised during GSC process may also be raised here for initial response of PP, as appropriate.

Summary of findings	CL	CAR	FAR
	4	3	3

Table 1. Remaining FAR from validation and/or previous verifications (Applicable for verification)

FAR ID	xx	Section no.	Date: DD/MM/YYYY
Description of FAR			
Project participant response		Date: : DD/MM/YYYY (please mention the date of your response to KBS)	
Please mention the response in this box.			
Documentation provided by project participant			
Please mention the document provided to KBS along with the above response.			
DOE assessment		Date: : DD/MM/YYYY	

Table 2.CL from this validation/verification

CL ID	CL 01	Section no.	1.1 of PD	Date: 03/10/2023
Description of CL				
In section 1.1 of the VCS PD, the capacity of the first project instance is stated as 28,000 TR. However, the plant capacity noted during the on-site assessment is 9,000 TR. Please clarify.				
Project participant response				Date: 07/10/2023
The VCS PD has been updated for 9000 TR capacity only. As this is grouped project additional capacity will be included during future verification as per the VCS program procedures. The emission reduction and other section of VCS PD are also updated accordingly.				
Documentation provided by project participant				
Updated VCS PD version 1.1				

DOE assessment	Date: 17/10/2023
The PP has updated the capacity of the project as 9000 TR capacity only. During the site assessment, the validation team noted that the PP has provided space for additional capacity i.e., 4000 TR of thermal storage space and 9000 TR with 3 chillers. The validation team also noted from Licence Application /26/ for the project activity that the PP has applied for a design capacity of 28,000 TR for the SeaWorld Cooling Plant. However only 9000 TR is installed at the time of validation. The validation team also noted from a Email communication dated 24-May-2023 from Department of Energy, addressing to the PP, TABREED /27/, confirming the Design capacity and Installed capacity. The validation team noted from the interview with the PP that capacity will be increased based on the demand from the end user. The validation team noted that since this is a grouped project additional capacity will be included during future verification as per the VCS program procedures. The emission reduction and other section of VCS PD are also updated accordingly to 9000 TR of installed capacity. Hence this CL 01 is closed.	

CL ID	CL 02	Section no.	3.5 of PD	Date: 03/10/2023
Description of CL				
In Section 3.5 of VCS PD, for the selection of the most plausible baseline scenario and demonstration of additionality, PP has applied the procedure described in Para 5.2.1 of the methodology: Step 1: Penetration and performance assessment, wherein following conditions are stated. 1. seasonal energy efficiency ratio (SEER) of the district cooling plant is higher than a benchmark SEER; and 2. The share of the district cooling technologies at the moment of the project registration is less than 20 per cent of all cooling technologies within the benchmark boundary in terms of cooling output; PP is requested to justify with evidences on how the two conditions are met.				
Project participant response				Date: 07/10/2023

The Project owner had appointed third party to carry out the Abu Dhabi cooling market analysis as per requirements of the methodology.

The detailed report is provided by the appointed party.

As per section 2.6 page 56 of the report, the cooling technologies penetration and SEER is plotted.

The following are the outcome:

1. The benchmark SEER is 2.26 while the project SEER is 4.03, therefore project SEER is higher than the benchmark and project meets this condition.
2. The district cooling penetration is 17.8% (~18%) which is less than 20%. Therefore, the project meets this condition.

VCS PD section 3.5 has been updated with all the reference document and information.

Documentation provided by project participant

1. Updated VCS PD version 1.1
2. Abu Dhabi cooling market report
3. Seaworld DC plant – daily cooling and power consumption data

DOE assessment

Date: 17/10/2023

The Project owner had appointed third party, Ms. Global Marker Insights to carry out the Abu Dhabi cooling market analysis as per requirements of the methodology.

In section 3.5 of the VCS PD, the benchmark boundary is chosen as Abu Dhabi City. However, Section 2.4 of the Baseline study report, 'Abu Dhabi Cooling Market' by Global Market Insights mentions that data sources from across the Abu Dhabi Emirates. Please clarify the scope of benchmark boundary.

Also, include the conditions to demonstrate baseline scenario and additionality as eligibility criteria for future addition of new project instances.

This CL 02 is open.

Project participant response

Date: 20/10/2023

As observed by the validation team, the data in the report has been used for the Abu Dhabi Emirate. The UAE is divided into seven administrative divisions (Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah), referred as emirates. Each emirates have its own rules and regulations and clear demarcation of boundary. As no clear administrative demarcation of city is used within the UAE therefore Abu Dhabi Emirate is considered as city for methodological requirement and benchmark boundary. As this is grouped project and therefore the boundary is limited to Abu Dhabi Emirate, United Arab Emirates.

The VCS PD has been updated to maintain the consistency.

Documentation provided by project participant

Updated VCS PD Version 1.1.

DOE assessment	Date: 31/10/2023
The PP has submitted a methodology deviation for selection of benchmark boundary in section 3.6 of VCS PD. The validation team noted that this deviation meets the requirements of the applicable VCS standard: The validation team noted that as per the applied Methodology, the default benchmark boundary is a city where the project activity is located. However, the validation team noted that the methodology does not provide any definition related to city. The validation team noted that the host country UAE is divided into seven administrative divisions¹⁰ (Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah), referred as 'Emirates'. Each Emirates have its own rules and regulations and clear demarcation of boundary. The PP has considered Abu Dhabi Emirate as benchmark boundary for demonstrating methodological requirements. As this is a grouped project and therefore the boundary is limited to Abu Dhabi Emirate, United Arab Emirates. The validation team noted that the Emirates have a high demand for air conditioning. From section 2.3.1.2 of the baseline study report by Global Market Insights Inc /47/, the validation team noted that the Prevalent elevated temperatures build unfavorable living conditions, which make air conditioning a necessity across the Emirates. Hence, the selection of benchmark boundary as Abu Dhabi Emirates is found to be reasonable. Hence this deviation is found to be acceptable. Hence, this CL 02 is closed.	

CL ID	CL 03	Section no.	Section 4 of PD	Date: 03/10/2023
Description of CL				
Provide reference for sources used for following assumptions in the Emission reduction calculation. - Load factor - 35 % - SEER of chillers with TES – 0.84 kW/TR - Average technical transmission and distribution losses – 7 %. - Amount of fresh water used – 703954 m³ – Based on the cooling tower capacity. - Emission factor associated with the production of freshwater, data provided by the supplier - Refrigerant charge – 31112 kg – Product data sheet - Annual leakage – 0.5 %				
Project participant response				Date: 07/10/2023

¹⁰ https://en.wikipedia.org/wiki/Politics_of_the_United_Arab_Emirates

Load Factor: This parameter is no more used. The annual data is calculated based on actual plant data. The daily data is submitted as '2-3 Seaworld DC plant – daily cooling and power consumption data'. The 133 days data is multiplied by (365/133) to extrapolate annual data.

SEER of the Chiller: 0.84 kW/TR is replaced with actual plant data as discussed in the Load Factor above.

Average technical transmission and distribution losses: This is updated to 8.18% based on Department of energy '2021 Annual Technical Report'. Page 31 for transmission losses (2.18%) and page 41 for distribution losses (6%). The report is available at <https://www.doe.gov.ae/-/media/Project/DOE/Department-Of-Energy/Media-Center-Publications/06-DOE-Annual-Technical-Report-2021-EN.pdf>

Furthermore, this parameter is included as a monitoring parameter.

Amount of fresh water used: This parameter is taken from the concept report page 28. The revised value is 1620.5 m³.

Emission factor for fresh water: The data is used at 11.5 kg CO₂/m³ (0.0115 ton CO₂/ton) based on publicly available information (page 80 Taqa sustainability report 2022). The data provided by supplier will be used in first monitoring period. This parameter is shifted to the monitoring parameters in the revised VCS PD.

Refrigerant charge: The value is from manufacturer's chiller data sheet. The chiller data sheets are submitted as '7 Product Data Sheet Chiller CSD and '8 Product Data Sheet Chiller VSD' where CSD is constant speed drive and VSD variable speed drive.

Annual leakage: This is estimated based on 'Chiller refrigerants: Minimizing carbon footprint, Johnson control (Chiller manufacture) page 5' Which has mentioned leakage of managed chillers is one-half percent (0.5%). The project chiller are manufactured by same company.

Documentation provided by project participant

1. Updated VCS PD version 1.1
2. Updated Emission reduction calculation
3. Seaworld DC plant – daily cooling and power consumption data
4. DOE Annual Technical Report 2021
5. Concept Design Report
6. Taqa sustainability report 2022
7. Product Data Sheet Chiller CSD
8. Product Data Sheet Chiller VSD
9. Chiller refrigerants: Minimizing carbon footprint, Johnson control (Chiller manufacture)

DOE assessment	Date: 17/10/2023
All assumptions and data used by the project participants are now listed in the PD, including their references and sources. All documentation used by the project participant as the basis for assumptions and source of data is correctly quoted and interpreted in the PD. All values used in the PD including GWPs are considered reasonable in the context of the proposed VCS grouped project activity. All estimates of the baseline GHG emissions can be replicated using the data and parameter values provided in the PD. Hence this CL 03 is closed.	

CL ID	CL 04	Section no.	4.1 of PD	Date: 03/10/2023
Description of CL				
In section 3.2 of the VCS PD, while demonstrating compliance to applicability criteria of the applied tool, 'Tool to calculate the emission factor for an electricity system (version 07)', it is stated 'the project replaces grid power supply and uses this tool to calculate the values of OM, BM and CM of this project'. PP is requested to provide the calculations for OM, BM and CM.				
Project participant response				Date: 07/10/2023
The data for calculation of OM, BM and CM is not available for UAE grid. The VCS PD has been updated to include this as methodology deviation where UAE grid emission factor is used from publicly available sources. The latest grid emission factor is used from UAE Energy profile document published by IRENA. All the reference updated in the revised VCS PD.				
Documentation provided by project participant				
1. Updated VCS PD version 1.1 2. Updated emission reduction calculation 3. UAE Energy Profile_IRENA				
DOE assessment				Date: 18/10/2023
The validation team noted that data for calculation of OM, BM and CM is not available for UAE grid. The VCS PD has been updated to include this as methodology deviation where UAE grid emission factor is used from publicly available sources. The latest grid emission factor is used from UAE Energy profile document published by IRENA. This CL is closed.				

Table 3.CAR from this validation/verification

CAR ID	CAR 01	Section no.	2.2 of PD	Date: 03/10/2023
Description of CAR				
In Section 2.2 Local Stakeholder consultation, describe the process for, and the outcomes from, the local stakeholder consultation conducted prior to validation. In doing so, include details on the following and provide supporting documents. a. The methods used for engaging local stakeholders (e.g., dates of announcements or meetings, periods during which input was sought). b. The methods used for documenting the outcomes of the local stakeholder consultation.				
Project participant response				Date: 07/10/2023
The stakeholder consultation was conducted on 19/09/2023. The invite for consultation were sent on 06/09/2023. The project owner has provided the project related information to all the stakeholder. During stakeholder meeting the presentation of made related with the project and stakeholders asked about the questions related with the SEER, benefits of the plants etc. Project owner has dedicated system for receiving any complaints in future as well which is included in the VCS PD. This will be documented.				
Documentation provided by project participant				
1. Updated VCS PD version 1.1 2. Stakeholder invitation and List of stakeholder invited 3. Photos of stakeholder meeting 4. Stakeholder presentation				
DOE assessment				Date: 18/10/2023
The PP has provided a description of local stakeholder consultation process in Section 2.2 of the VCS PD and included the following. i. Procedures for engaging local stakeholders; ii. Type of stakeholders and number of stakeholders that attended the meeting and; iii. Mechanism for on-going communication with local stakeholders. The validation team noted that the PP has adopted established procedures for carrying out effective stakeholder consultation meeting, Hence, CAR 01 is closed.				

CAR ID	CAR 02	Section no.	5.1	Date: 03/10/2023
Description of CAR				
In section 5.1, Data and Parameters Available at Validation, Not all information is filled/provided for some of the parameters to complete the table. Example: Parameter 'Building type'				
Project participant response				Date: 07/10/2023
VCS PD is updated to include all the information of the parameter as required in the methodology.				
Documentation provided by project participant				
Updated VCS PD version 1.1				
DOE assessment				Date: 18/10/2023
For the parameter, 'Building type', The project will be providing the cooling demand to only new buildings. This is confirmed through interview with the PP. Hence, the building types is categorized as new. For the parameter, 'Baseline cooling technologies', following is provided. The validation team noted that data collection on baseline cooling technologies has been done by third party, Global Market Insights based on various data sources, including Abu Dhabi Department of Energy (DoE), Abu Dhabi Industrial Strategy (ADIS), Abu Dhabi Department of Economic Development (ADDED), Abu Dhabi Industrial Development Bureau (IDB). Hence, the values determined are found acceptable. The validation team noted that adequate information is provided for the ex-ante parameters. Hence this CAR 02 is closed.				

CAR ID	CAR 03	Section no.	5.2	Date: 03/10/2023
Description of CAR				
In section 5.2: Following corrective actions are requested. <i>Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).</i> <i>Specify measurement and recording frequency</i> <i>Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.</i> <i>Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures</i>				
Project participant response				Date: 07/10/2023

The section 5.2 is updated with serial number/reference where relevant including calibration and monitoring frequency.

Documentation provided by project participant

Updated VCS PD Version 1.1.

DOE assessment

Date: 18/10/2023

The Validation team reviewed the revised PD and noted that the overall monitoring plan complies with the requirements of the methodology AM0117 /40/; the monitoring parameters described in section 5.2 of the VCS PD and their monitoring arrangements described in the monitoring plan are feasible within the project design.

Hence this CAR 03 is closed.

Table 4.FAR from this verification

FAR ID	01	Section no.	3.1 of this report	Date: 15/11/2023
Description of FAR				
The project ownership for each project instance, (satisfying the requirements of Section 3.7 of VCS Standard version 4.5), is to be verified from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).				
Project participant response				Date: DD/MM/YYYY
NA.				
Documentation provided by project participant				
NA.				
DOE assessment				Date: DD/MM/YYYY
NA.				

FAR ID	02	Section no.	3.3.5 of this report	Date: 15/11/2023
Description of FAR				
Benchmark will be updated every 3 years after registration of the project, as per the requirements of para 36 of applied methodology.				

Project participant response	Date: DD/MM/YYYY
NA.	
Documentation provided by project participant	
NA.	
DOE assessment	Date: DD/MM/YYYY
NA.	

FAR ID	03	Section no.	3.3.8 of this report	Date: 15/11/2023
Description of FAR				
The verification team is requested to verify the value of Global Warming potential of refrigerant for future commitment periods.				
Project participant response				Date: DD/MM/YYYY
NA.				
Documentation provided by project participant				
NA.				
DOE assessment				Date: DD/MM/YYYY
NA.				